Hold On to Your Cash:

CEO Speech Uncertainty and Financing Conservatism

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May 4, 2026

Abstract

We ask whether CEO speech uncertainty during earnings calls predicts firm precautionary cash holdings. Using the speech-uncertainty method of Dzielínski, Wagner, and Zeckhauser (2021), we extend their decomposition from market reactions, analyst behavior, firm performance, and governance outcomes to firm financing-policy outcomes on a panel of U.S. earnings calls from 2002 to 2018. Within-firm increases in the residual Q&A component of CEO speech uncertainty load positively on the cash-to-assets ratio in twelve of twelve specifications under one-tailed tests at the ten-percent level. The cash response amplifies in the cross-section under both financing-friction stress and cash-flow-volatility stress, consistent with a single precautionary channel triggered by two distinct stress conditions. External uncertainty proxies covary with the speech measure in the predicted directions; post-call bid-ask spreads widen and the SEC subsequently issues conference-call comment letters at higher rates following calls with elevated speech uncertainty. Reverse-causality concerns are addressed via cross-sectional modus tollens on the two precautionary triggers; we acknowledge as a limitation that no exogenous-shock identification is pursued.

I Introduction

Earnings calls are a high-frequency setting where CEOs describe firm conditions to outsiders in their own words. We ask whether CEO speech uncertainty during these calls predicts firm precautionary cash holdings. Our measurement is built on the CEO speech-uncertainty method of Dzielínski et al. (2021), which we extend from market reactions, analyst behavior, firm performance, and governance outcomes to firm financing-policy outcomes.

We document that within-firm increases in CEO speech uncertainty during earnings calls predict higher precautionary cash holdings, identifying a real-time signal of firm financing-policy adjustment grounded in the precautionary cash motive (Opler, Pinkowitz, Stulz, & Williamson, 1999; Bates, Kahle, & Stulz, 2009). The relation is positive in twelve of twelve specifications (six contemporaneous and six one-quarter-ahead) under one-tailed tests at the ten-percent level. The cash response amplifies under both financing-friction stress and cash-flow-volatility stress, consistent with a single precautionary channel triggered by two distinct stress conditions (Almeida, Campello, & Weisbach, 2004; Han & Qiu, 2007).

External uncertainty proxies—political risk, economic policy uncertainty, and product-market competition—covary with the speech-uncertainty measure in the predicted directions, supporting the measure's construct validity. Post-call bid-ask spreads widen and SEC conference-call comment letters reference the same speech signal, corroborating the precautionary interpretation through market information-asymmetry and disclosure-insufficiency channels.

Reverse-causality concerns are addressed via cross-sectional modus tollens on the two precautionary triggers, and we acknowledge as a limitation that no exogenous-shock or instrumental-variable identification is pursued.

Our main contribution is to establish CEO speech uncertainty during earnings calls as a real-time signal of firm precautionary cash holdings, extending the method beyond its original scope of market reactions, analyst behavior, firm performance, and governance outcomes. The remainder of the paper proceeds as follows. Section II develops the conceptual framework, hypotheses, and empirical strategy. Section III presents the main empirical analyses—cash holdings, the two precautionary triggers, and the endogeneity treatment. Section IV extends to outsider-reaction tests. Section V concludes with limitations and directions for future work.

II Conceptual Framework and Empirical Strategy

II.1 Conceptual Framework

Firms hold cash to insure against future funding shortfalls. The precautionary cash motive predicts that firms with riskier cash flows or poor access to external capital hold more cash (Opler et al., 1999; Bates et al., 2009). The motive operates through a single mechanism: cash buffers allow firms to fund profitable investment without recourse to costly external capital, mitigating underinvestment when internal cash flow falls short of investment needs. Firms most exposed to this risk—those facing the largest gap between expected internal cash flow and expected required investment—optimally hold the largest buffers.

Two stress conditions amplify the precautionary response. The first is financing-friction stress, in which limited debt-market access leads firms to accumulate cash to safeguard future investment (Almeida et al., 2004); the second is cash-flow-volatility stress, in which volatile operating cash flows lead firms to hold larger buffers to smooth funding gaps (Han & Qiu, 2007). The two stresses are conceptually distinct: one constrains the supply of external capital, the other increases the variance of internal capital. Both trigger the same precautionary response, generating two testable amplifications of the cash response under stress.

II.2 Hypothesis Development

We formalize three hypotheses linking CEO speech uncertainty during earnings calls to firm precautionary cash holdings. The main hypothesis predicts a positive association between speech uncertainty and cash; two amplification hypotheses predict that the association strengthens under each of the two stress triggers identified in Section II.

Speech uncertainty is a candidate real-time signal of the precautionary motive because earnings-call language varies with managerial perceptions of firm conditions (Dzielínski et al., 2021). When CEOs perceive heightened firm-level uncertainty, the precautionary mechanism predicts that firms hold more cash to insure against future funding shortfalls (Opler et al., 1999; Bates et al., 2009).

Hypothesis 1. *An increase in CEO speech uncertainty during an earnings call is associated with a higher cash-to-assets ratio at the firm.*

The two stress conditions of Section II predict heterogeneous amplification of H1. Financing-friction stress operates through the supply of external capital: rated firms can issue debt at low cost, while unrated firms face binding external-finance constraints and accumulate cash to safeguard future investment (Faulkender & Petersen, 2006; Almeida et al., 2004). Cash-flow-volatility stress operates through the variance of internal capital: firms with volatile operating cash flows hold larger buffers to smooth funding gaps (Han & Qiu, 2007).

Hypothesis 1a. *The association between CEO speech uncertainty and cash holdings is stronger for firms facing financing-friction stress (unrated firms) than for firms with debt-market access (rated firms).*

Hypothesis 1b. *The association between CEO speech uncertainty and cash holdings is stronger for firms facing cash-flow-volatility stress (high-volatility firms) than for firms with stable cash flows (low-volatility firms).*

II.3 Estimation of Main Variables

We measure CEO speech uncertainty during earnings calls following Dzielínski et al. (2021). We construct two independent variables of interest from the percentage of uncertainty-words in CEO speech segments: UncResCEO is extracted from the unscripted Q&A segment via the DWZ decomposition, and UncPreCEO is the raw percentage in the scripted presentation segment. We also include ClarityCEO, the persistent CEO-trait component of the same DWZ decomposition, as a control to absorb between-CEO variation in habitual speech style.

UncResCEO is the residual from a regression of Q&A-segment uncertainty on a CEO entity fixed effect and call-level controls (DWZ Eq. 4); it captures within-CEO variation in Q&A uncertainty orthogonal to the CEO's habitual baseline. ClarityCEO is the negative of the CEO entity fixed effect from the same regression (DWZ Eq. 5); it captures the CEO's habitual baseline level of uncertainty-language and enters our specifications as a control absorbing persistent CEO-trait variation. UncPreCEO is the raw percentage of uncertainty-words in the CEO's scripted opening, typically prepared in advance and vetted by investor-relations staff before the call.

Our use of UncPreCEO as an independent variable extends the original method. Dzielínski et al. (2021) use the presentation-segment measure as a control variable in their Q&A residual extraction. We treat UncPreCEO as an independent variable because it may carry an informative signal about firm conditions. We disclose this departure and report results separately for each of the three independent variables in Section III.

II.4 Methodology

We estimate panel regressions of cash holdings on the speech-uncertainty independent variables of interest (UncResCEO and UncPreCEO), the ClarityCEO control, and a vector of firm-level controls. The dependent variable is the cash-to-assets ratio, measured contemporaneously (Cash_t) and at a one-quarter horizon (Cash_{t+1}).

For each dependent-variable horizon, we report a six-cell fixed-effects ladder: (i) industry fixed effects with base controls; (ii) firm fixed effects with base controls; (iii)–(iv) the same two specifications with extended controls; and (v)–(vi) the two extended-control specifications with calendar year-quarter fixed effects added. The base control set follows Bates et al. (2009) (firm size, leverage, market-to-book, ROA, capital expenditure, dividend payer, and operating-cash-flow volatility); the extended set adds sales growth, R&D intensity, cash flow, and stock-return volatility. The lagged dependent variable is included as a base control in all specifications to absorb cash-policy persistence.

Standard errors are clustered at the firm level. Tests of the main hypothesis are one-tailed in the predicted direction; tests of the amplification hypotheses are one-tailed in the predicted direction of the interaction sign.

II.5 Specification and Measurement of Key Constructs

Before turning to the cash analyses, we evaluate the construct validity of the speech-uncertainty measure by testing whether it covaries with established external proxies for firm-relevant uncertainty. If the measure captures uncertainty information rather than mechanical text-frequency artifacts, it should move with proxies that have been validated as uncertainty signals in prior literature.

We use four external proxies: firm-level political-risk exposure measured from earnings-call discussion of political topics (Hassan, Hollander, van Lent, & Tahoun, 2020); aggregate U.S. economic policy uncertainty constructed from newspaper coverage (Baker, Bloom, & Davis, 2016); global economic policy uncertainty as a GDP-weighted average across countries (Davis, 2016); and product-market competition intensity from text-based industry-network similarity (Hoberg & Phillips, 2016).

We expect UncResCEO and UncPreCEO—the within-CEO time-varying components of the speech-uncertainty measure—to load positively on the four proxies. ClarityCEO is a CEO-level fixed effect and is excluded from this test. The construct-validity results are reported alongside the main empirical analyses.

III Main Empirical Analyses

III.A Data, Sample, and Variable Construction

Our sample covers earnings-call observations of U.S. public firms over the period 2002 through 2018 at the firm-quarter frequency. The full call panel comprises 112,968 firm-quarter observations across 2,429 distinct firms. We obtain financial variables from Compustat and earnings-call text features from a vendor-supplied transcript repository.

We exclude firms in financial services and utilities (Fama-French 12 industries 11 and 8). The estimation sample also requires a non-missing lagged dependent variable and is restricted to firms with multiple observations after firm-level fixed-effect singleton dropping. The resulting sample comprises approximately 43,300 firm-quarter observations across 1,376 firms in the contemporaneous specifications and approximately 41,100 firm-quarter observations across 1,321 firms in the lead specifications.

Section II describes the construction of the two CEO speech-uncertainty independent variables of interest (UncResCEO and UncPreCEO) and the ClarityCEO control. The cash-to-assets ratio (Cash_t, Cash_{t+1}) is cash and short-term investments scaled by total assets. Variable definitions are provided in the appendix; summary statistics are reported in Table 1.

III.B Cash Holdings (HC)

Table 2 reports the main estimation results for cash holdings on the speech-uncertainty independent variables of interest. Across the twelve specifications, UncResCEO loads positively on cash holdings in every cell, with one-tailed p-values below the ten-percent level. UncPreCEO is positive but statistically indistinguishable from zero across all twelve specifications.

As a robustness check, Table 3 replicates the main results using a quarterly-expanding-window construction of UncResCEO that eliminates look-ahead bias by re-estimating the speech-uncertainty decomposition each quarter from data available through that quarter only. UncResCEO remains positive and significant in twelve of twelve specifications; the UncPreCEO pattern is preserved.

III.C Constraint Amplification (HFC)

H1a predicts that the cash response amplifies under financing-friction stress, with constrained firms (proxied by an unrated debt status) showing a stronger relation between speech uncertainty and cash holdings than rated firms (Almeida et al., 2004). Table 4 reports estimates of the H1 specification with the addition of an Unrated dummy and its interactions with UncResCEO and UncPreCEO. ClarityCEO is not interacted with Unrated, since the constructed CEO trait does not separate cleanly from cross-sectional firm constraints.

The UncResCEO × Unrated interaction loads positively across all eight interaction specifications. The interaction is significant at the ten-percent level under one-tailed tests in the lead-cash specifications with industry fixed effects (columns 13 and 15) but is not significant in the contemporaneous specifications or in the firm-fixed-effects cells. The UncPreCEO × Unrated interaction is direction-consistent in all eight specifications and is significant in one cell (column 7).

The amplification pattern is direction-consistent in every cell. Financing friction amplifies the cash response in the cross-section, with statistical detection in the lead-cash specifications under industry fixed effects. The within-firm variation in the Unrated dummy is sparse—firms rarely transition between rated and unrated status within the sample period—so the interaction coefficient is identified primarily off cross-firm variation in financing friction. Table 5 replicates the analysis using the quarterly-expanding-window construction; the pattern is preserved.

III.D Cash-Flow-Volatility Amplification

H1b predicts that the cash response amplifies under cash-flow-volatility stress, with high-volatility firms showing a stronger relation between speech uncertainty and cash holdings than low-volatility firms (Han & Qiu, 2007). Cash-flow volatility (CFvol) is constructed as the coefficient of variation of operating cash flow over a sixteen-quarter rolling window, following Han and Qiu (2007). The HighCFvol indicator equals one when a firm-quarter is at or above the Fama-French 12 industry-year median CFvol. Table 6 reports estimates of the H1 specification with the addition of HighCFvol and its interactions with UncResCEO and UncPreCEO.

The UncPreCEO × HighCFvol interaction loads positively in all eight specifications and is significant at the ten-percent level under one-tailed tests in four of eight cells—all four with industry fixed effects, across both the contemporaneous and lead horizons. The UncResCEO × HighCFvol interaction is direction-consistent in all eight cells and significant in two of eight (lead-cash specifications with industry fixed effects). The interaction effects are absorbed under firm fixed effects, mirroring the pattern in Section III.

Cash-flow-volatility stress amplifies the cash response in the cross-section, with statistical detection across both horizons under industry fixed effects. The H1b specification adopts the cash-flow-volatility construction of Han and Qiu (2007) as a moderator on the speech–cash relation. Together with the financing-friction amplification of Section III, the two interactions cover both stress dimensions of the precautionary cash motive.

III.E Endogeneity

The most consequential endogeneity concern in our setting is reverse causality: cash policy could shape CEO speech rather than the other way around. Within-firm fixed effects and lagged-DV controls absorb time-invariant firm heterogeneity and policy persistence, but neither rules out the reverse arrow at the within-firm time-series level. We address this concern through cross-sectional modus tollens on the two precautionary triggers identified in the conceptual framework: if the forward direction (speech uncertainty → cash) operates through the precautionary mechanism, the cash response should amplify under stress conditions that are themselves cross-sectionally exogenous to within-firm cash-policy choices.

Sections III and III provide such evidence. The cash response amplifies in the cross-section under both financing-friction stress (rated versus unrated debt status) and cash-flow-volatility stress (high versus low CFvol). Neither moderator is plausibly endogenous to within-firm CEO speech: rated/unrated status is determined primarily by firm size and reputation external to quarterly speech variation, and CFvol is constructed from a sixteen-quarter trailing window, predating the contemporaneous speech measure. The observed amplification pattern is consistent with the forward direction and harder to reconcile with a reverse-causation alternative.

We acknowledge as a limitation that no exogenous-shock or instrumental-variable identification is pursued. We considered and dropped six candidate designs during the project: a CEO sudden-death difference-in-differences, a first-difference reformulation of the Dzielinski-Wagner-Zeckhauser specification, a Lewbel heteroskedasticity-based instrument, weather-shock instruments, the American Jobs Creation Act repatriation experiment, and the Almeida-Campello-Laranjeira-Weisbenner long-term-debt-maturity difference-in-differences. Each failed for design-specific reasons including mechanism mismatch, parallel-trends violations, dual-channel contamination, and absence of a documented cash response to the candidate shock.

The most promising remaining candidate is the firm-level political-risk text measure of Hassan et al. (2020) interacted with major political shocks. We flag this direction as future work but do not pursue it in the present paper.

IV Additional Analyses

IV.A Market Information-Asymmetry Channel: Post-Call Bid-Ask Spread

The market information-asymmetry channel predicts that post-call bid-ask spreads widen following CEO speeches with higher uncertainty, as outside investors price-protect against the informed-trading risk that the speech signal makes salient. We measure the post-call response on a twenty-five-trading-day window beginning the call day. Table 12 reports estimates of the speech-uncertainty independent variables on the twenty-five-day spread level, in basis points, under the four-step fixed-effects ladder and the market-microstructure controls (stock price, share turnover, daily return volatility, absolute earnings surprise) that the spread literature requires.

UncPreCEO loads positively on the contemporaneous-quarter spread in all six contemporaneous specifications and is significant at the ten-percent level under one-tailed tests in four of six cells. The pattern preserves under firm fixed effects across all three control specifications; under industry fixed effects, significance is preserved with base controls but is absorbed once the market-microstructure controls are added (columns 3 and 5). The lead-quarter specifications are null. UncResCEO loads null across both horizons.

The contemporaneous market response is consistent with the precautionary interpretation of the speech signal: outside investors impound the signal into liquidity provision in the same quarter, widening spreads alongside the within-firm cash response documented in Section III. The presentation-segment component of CEO speech, rather than the residual Q&A component, carries the market-relevant signal at this horizon.

IV.B Disclosure-Insufficiency Channel: Conference-Call Comment Letter (CCCL)

The disclosure-insufficiency channel predicts that the SEC issues a Conference-Call Comment Letter (CCCL) when an earnings call surfaces disclosure problems that warrant follow-up review. Following Lerman, Steffen, and Zhang (2026), who introduce and validate the CCCL as a regulatory-scrutiny indicator referencing the earnings call, we measure CCCL receipt as a binary outcome and estimate a linear probability model. Table 13 reports estimates of the speech-uncertainty independent variables on CCCL receipt under the four-step fixed-effects ladder.

UncPreCEO loads positively on CCCL receipt in all six specifications and is significant at the ten-percent level under one-tailed tests in four of six cells. Significance is preserved in three of the four specifications with extended controls (columns 3, 4, and 6) and is absorbed only under industry-year-quarter fixed effects with extended controls (column 5). UncResCEO loads null across all six specifications.

The pattern is consistent with the disclosure-insufficiency interpretation: presentation-segment uncertainty during the earnings call associates with a higher probability that the SEC subsequently issues a comment letter referring to the call. As in Section IV, the presentation-segment component of CEO speech, rather than the residual Q&A component, carries the regulator-relevant signal.

V Conclusion

V.1 Summary

This paper documents that within-firm increases in CEO speech uncertainty during earnings calls predict higher precautionary cash holdings. The residual

Q&A component of CEO speech-uncertainty (UncResCEO) loads positively on the cash-to-assets ratio in twelve of twelve specifications under one-tailed tests at the ten-percent level. The cash response amplifies under both financing-friction stress and cash-flow-volatility stress, consistent with a single precautionary channel triggered by two distinct stress conditions (Almeida et al., 2004; Han & Qiu, 2007).

The contribution is to establish CEO speech uncertainty as a real-time signal of firm precautionary cash holdings, extending the speech-uncertainty method of Dzielínski et al. (2021) beyond its original scope—market reactions, analyst behavior, firm performance, and governance outcomes—to firm financing-policy outcomes. Outsider reaction corroborates the precautionary interpretation through two channels: post-call bid-ask spreads widen contemporaneously, and the SEC subsequently issues Conference-Call Comment Letters at higher rates following calls with elevated speech uncertainty.

Reverse-causality concerns are addressed through cross-sectional modus tollens on the two precautionary triggers identified in the conceptual framework. We acknowledge as a limitation that no exogenous-shock or instrumental-variable identification is pursued; Section V discusses this and the candidate designs we considered.

V.2 Limitations

The primary limitation of our identification strategy is the absence of an exogenous-shock or instrumental-variable defense of the speech-uncertainty → cash direction. As discussed in Section III, we identify the forward direction through cross-sectional modus tollens on two precautionary triggers; six candidate exogenous-shock and instrumental-variable designs were considered and dropped during the project. The CEO sudden-death difference-in-differences was dropped for mechanism mismatch: the shock affects multiple firm channels simultaneously rather than a narrow speech channel. The first-difference reformulation of the Dzielínski et al. (2021) specification was dropped because it addresses a time-invariant firm-trait threat that within-firm fixed effects in our main panel already absorb. The Lewbel heteroskedasticity-based instrument failed the Wu-Hausman test for OLS-inconsistency. Weather-shock instruments lack a documented cash-holdings response to identify a first stage. The 2004 American Jobs Creation Act repatriation experiment was dropped because the load-bearing public-finance evidence on this shock reports no cash-holdings response post-shock. The long-term-debt-maturity difference-in-differences anchored on the 2007 credit-crisis maturing-debt cohort was dropped for dual-channel contamination: the shock pushes cash in the opposite direction of the hypothesized speech channel via a separate forced-deployment mechanism.

Any future exogenous-shock identification compatible with our forward-direction hypothesis must satisfy a strict requirement set: exogeneity to firm cash policy, a narrow speech channel without parallel direct-on-cash mechanisms, direction-compatibility on any non-speech channels, a first stage on speech, cross-sectional intensity variation, parallel trends on both cash and speech, a top-tier journal anchor, and an adequate treated-firm count. The strongest a-priori candidate we are aware of is discussed in Section V.

Two methodological-extension disclosures bear noting. We use UncPreCEO as an independent variable, whereas Dzielínski et al. (2021) treat the presentation-segment measure as a control in their residual-extraction step. We extend the speech-uncertainty method to firm financing-policy outcomes; Dzielínski et al. (2021) themselves report null effects of UncResCEO and UncPreCEO on the market-reaction outcomes that motivated the original method, and our positive cash-side findings should be read in light of that scope difference.

V.3 Future Work

Among the designs evaluated against the requirement set in Section V, the strongest a-priori candidate is the firm-level political-risk text measure of Hassan et al. (2020) interacted with a major political shock such as the 2016 U.S. presidential election. The cross-sectional intensity variation is provided by pre-shock firm-level political-risk exposure constructed from earnings-call language; the time-series identifying variation is provided by the shock itself; and the speech-uncertainty mechanism is direct, since the political-risk measure is itself constructed from earnings-call text.

We flag this design as a research direction rather than pursue it in the present paper. It has not yet been verified against the full requirement set; in particular, the parallel-trends condition on cash holdings and on speech uncertainty requires confirmation, and the dual-channel risk—political shocks plausibly affect cash holdings through non-speech channels including investment-policy and tax-policy uncertainty—requires careful direction-compatibility analysis. We leave these checks to future work.

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Table 1: Summary Statistics

Variable	Unit	N	Mean	SD	Min	P25	Median	P75	Max
A. Independent Variables									
ClarityCEO	C	61,626	0.2143	0.2129	-0.7717	0.0952	0.2365	0.3629	0.8614
ClarityCEO_QtrExp	C	25,520	0.2011	0.2281	-1.0232	0.0713	0.2256	0.3589	0.8694
GEPU_log	C	88,205	4.6872	0.3672	3.8626	4.3797	4.7238	4.9666	5.5106
PRisk	C	85,995	99.6187	146.5189	0.0000	17.0778	53.3729	120.1973	1192.4308
PRisk_lag2	C	80,627	99.9614	147.2392	0.0000	17.1702	53.4446	120.2666	1192.4308
TotalSimilarity	F	22,403	3.4277	5.6522	1.0000	1.1246	1.5865	3.0903	80.1876
US_EPU_log	C	88,205	4.7123	0.3661	3.8018	4.4805	4.6891	4.9625	5.6478
UncPreCEO	C	71,727	0.6602	0.3917	0.0000	0.3810	0.6001	0.8723	2.2432
UncResCEO	C	44,900	-0.0000	0.3010	-1.5108	-0.1846	-0.0180	0.1650	1.7794
UncResCEO_QtrExp	C	25,520	-0.0057	0.3208	-1.6084	-0.2028	-0.0210	0.1707	1.7402
Unrated	C	79,487	0.5159	0.4997	0.0000	0.0000	1.0000	1.0000	1.0000
B. Dependent Variables									
BGTLevel_Spread	C	87,018	0.0017	0.0026	0.0001	0.0004	0.0009	0.0017	0.0164
BGTLevel_Spread_lead1	C	82,843	0.0016	0.0024	0.0001	0.0004	0.0009	0.0016	0.0164
CCCL	C	88,205	0.0032	0.0563	0.0000	0.0000	0.0000	0.0000	1.0000
CashRatio	C	87,990	0.1708	0.1806	0.0000	0.0368	0.1047	0.2435	0.9938
CashRatio_lead	C	82,236	0.1715	0.1748	0.0000	0.0407	0.1112	0.2447	0.9938
C. Firm Controls									
AbsSurpDec	C	61,544	2.9300	1.4255	0.0000	2.0000	3.0000	4.0000	5.0000
Capex	C	87,627	0.0487	0.0536	0.0000	0.0176	0.0324	0.0592	0.5203
CashFlowAt	C	87,670	0.1019	0.1147	-3.0324	0.0608	0.1032	0.1526	0.4874
DailyVola	C	82,599	36.7556	21.2437	9.4388	22.9709	31.2762	43.8148	211.9160
DivDummy	C	88,134	0.4755	0.4994	0.0000	0.0000	0.0000	1.0000	1.0000
EarnVol	C	87,440	0.0701	0.1907	0.0006	0.0163	0.0315	0.0695	19.5101
FirmMat	C	87,571	-0.0432	2.1826	-317.5714	-0.0078	0.2191	0.4195	0.9252
Leverage	C	87,994	0.2330	0.2240	0.0000	0.0470	0.2065	0.3455	3.9453
NegCall	C	86,858	0.9210	0.3024	0.0000	0.7038	0.8798	1.0943	2.3901
RDSales	C	87,647	0.1047	0.9342	0.0000	0.0000	0.0048	0.0636	39.7688
ROA	C	87,552	0.0386	0.1330	-3.9826	0.0146	0.0525	0.0925	0.5490
SalesGrowth	C	87,506	0.1173	0.3849	-1.0000	-0.0043	0.0745	0.1762	10.3051
StockPrice	C	87,188	36.8864	32.8119	1.6100	14.8975	28.2400	48.1700	192.0384
TobinsQ	C	87,363	2.1363	1.5197	0.3847	1.2681	1.6791	2.4362	35.6144
Turnover	C	87,188	0.0260	0.0299	0.0007	0.0083	0.0160	0.0310	0.1697
UncQue	C	81,031	1.4398	0.4831	0.0000	1.1216	1.4108	1.7268	3.5945
lnAssets	C	87,994	7.4710	1.6607	-0.5692	6.3008	7.3763	8.5436	12.4592
sCFO	C	83,662	0.0619	0.2233	0.0014	0.0236	0.0395	0.0664	32.4035

Notes: Summary statistics for variables used in the hypothesis tests. Sample period: 2002–2018. Anchor sample (H1, Main): 88,205 earnings-call observations across 1,884 firms. Unit column: C = call-level, F = firm-year. N varies by row because each variable is summarized on its anchor panel's Main sample (complete cases per variable); per-suite samples differ due to lags, control availability, and merges.

Table 2: H1 CEO 3-IV Decomp: ClarityCEO + UncResCEO (DWZ Full) + UncPreCEO

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	CashRatio						CashRatio_lead					
CEO Clarity (DWZ)	**-0.0046***	**-0.0071***	**-0.0058****	-0.0054	**-0.0056****	-0.0052	-0.0052	**-0.0170****	**-0.0068***	**-0.0162****	**-0.0069***	**-0.0161****
	(0.0033)	(0.0051)	(0.0033)	(0.0048)	(0.0033)	(0.0047)	(0.0052)	(0.0095)	(0.0051)	(0.0090)	(0.0051)	(0.0090)
CEO Residual Unc. (DWZ)	**0.0021****	**0.0019****	**0.0020****	**0.0016***	**0.0021****	**0.0016***	**0.0028****	**0.0023****	**0.0026****	**0.0020***	**0.0025****	**0.0019***
	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0014)	(0.0013)	(0.0014)	(0.0013)	(0.0014)	(0.0013)
CEO Pres Uncertainty	0.0002	0.0009	0.0003	0.0006	0.0004	0.0007	-0.0003	0.0008	-0.0005	0.0004	-0.0007	0.0002
	(0.0011)	(0.0011)	(0.0011)	(0.0011)	(0.0011)	(0.0011)	(0.0021)	(0.0017)	(0.0021)	(0.0017)	(0.0021)	(0.0017)
Leverage	**-0.0177*****	**-0.0355*****	**-0.0130*****	**-0.0218*****	**-0.0132*****	**-0.0216*****	**-0.0327*****	**-0.0279****	**-0.0274*****	-0.0159	**-0.0275*****	-0.0156
	(0.0048)	(0.0084)	(0.0036)	(0.0082)	(0.0037)	(0.0082)	(0.0120)	(0.0121)	(0.0104)	(0.0118)	(0.0105)	(0.0118)
lnAssets	**-0.0015*****	**-0.0177*****	**-0.0017*****	**-0.0175*****	**-0.0014*****	**-0.0170*****	**-0.0024****	**-0.0268*****	**-0.0020****	**-0.0265*****	**-0.0020****	**-0.0265*****
	(0.0006)	(0.0026)	(0.0005)	(0.0026)	(0.0005)	(0.0026)	(0.0010)	(0.0042)	(0.0009)	(0.0042)	(0.0010)	(0.0042)
TobinsQ	**0.0058*****	**0.0048*****	**0.0042*****	**0.0037*****	**0.0040*****	**0.0034*****	**0.0085*****	0.0025	**0.0056*****	0.0009	**0.0056*****	0.0010
	(0.0006)	(0.0009)	(0.0007)	(0.0010)	(0.0007)	(0.0010)	(0.0012)	(0.0015)	(0.0012)	(0.0015)	(0.0012)	(0.0016)
ROA	-0.0144	**0.0214****	**-0.0679*****	-0.0195	**-0.0630*****	-0.0171	**-0.0762*****	**-0.0352***	**-0.1427*****	**-0.0790*****	**-0.1378*****	**-0.0769*****
	(0.0093)	(0.0105)	(0.0113)	(0.0122)	(0.0114)	(0.0122)	(0.0161)	(0.0185)	(0.0173)	(0.0194)	(0.0174)	(0.0196)
Capex	**-0.1117*****	**-0.2995*****	**-0.1596*****	**-0.3074*****	**-0.1564*****	**-0.2993*****	**-0.1613*****	**-0.3281*****	**-0.2433*****	**-0.3415*****	**-0.2411*****	**-0.3430*****
	(0.0128)	(0.0234)	(0.0149)	(0.0236)	(0.0148)	(0.0234)	(0.0184)	(0.0344)	(0.0216)	(0.0345)	(0.0216)	(0.0346)
DivDummy	**-0.0041*****	-0.0028	**-0.0054*****	**-0.0041***	**-0.0049*****	-0.0036	**-0.0051****	-0.0068	**-0.0056****	**-0.0076***	**-0.0054****	**-0.0077***
	(0.0013)	(0.0023)	(0.0013)	(0.0022)	(0.0013)	(0.0022)	(0.0023)	(0.0045)	(0.0023)	(0.0044)	(0.0023)	(0.0044)
sCFO	**0.0066*****	-0.0022	**0.0079*****	-0.0008	**0.0074*****	-0.0009	**0.0137****	0.0027	**0.0137****	0.0035	**0.0134****	0.0034
	(0.0020)	(0.0014)	(0.0030)	(0.0014)	(0.0028)	(0.0014)	(0.0057)	(0.0025)	(0.0062)	(0.0027)	(0.0061)	(0.0027)
Lagged_DV	**0.8558*****	**0.6146*****	**0.8612*****	**0.6322*****	**0.8617*****	**0.6337*****	**0.7214*****	**0.2209*****	**0.7227*****	**0.2377*****	**0.7231*****	**0.2385*****
	(0.0076)	(0.0133)	(0.0075)	(0.0130)	(0.0075)	(0.0130)	(0.0138)	(0.0201)	(0.0133)	(0.0201)	(0.0133)	(0.0201)
SalesGrowth			**-0.0178*****	**-0.0274*****	**-0.0169*****	**-0.0266*****			**-0.0156****	**-0.0171*****	**-0.0144****	**-0.0163*****
			(0.0045)	(0.0052)	(0.0044)	(0.0052)			(0.0061)	(0.0050)	(0.0060)	(0.0050)
RDSales			**0.0062*****	**-0.0032*****	**0.0062*****	**-0.0031****			**0.0137*****	-0.0014	**0.0138*****	-0.0012
			(0.0017)	(0.0012)	(0.0017)	(0.0012)			(0.0024)	(0.0019)	(0.0024)	(0.0019)
CashFlowAt			**0.1292*****	**0.1589*****	**0.1278*****	**0.1576*****			**0.1982*****	**0.1687*****	**0.1944*****	**0.1660*****
			(0.0162)	(0.0173)	(0.0162)	(0.0173)			(0.0236)	(0.0249)	(0.0237)	(0.0250)
DailyVola			0.0000	**-0.0001*****	**0.0001*****	**-0.0001***			**0.0003*****	**0.0001*****	**0.0003*****	0.0001
			(0.0000)	(0.0000)	(0.0000)	(0.0000)			(0.0000)	(0.0000)	(0.0001)	(0.0001)
Extended Controls			Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes
Industry FE	Yes		Yes		Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes		Yes		Yes
Year FE	Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes		
Year-Quarter FE					Yes	Yes					Yes	Yes
N	43,333	43,333	43,316	43,316	43,316	43,316	41,122	41,122	41,108	41,108	41,108	41,108
R^2	0.825	0.431	0.829	0.449	0.830	0.449	0.659	0.092	0.667	0.108	0.668	0.108
Adj. R^2	0.825	0.412	0.829	0.430	0.830	0.430	0.658	0.062	0.667	0.078	0.667	0.077

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). R^2 includes absorbed fixed effects (not within-R^2).

Table 3: H1 CEO 3-IV Decomp: ClarityCEO + UncResCEO (DWZ Qtr-Exp, no-look-ahead) + UncPreCEO

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	CashRatio						CashRatio_lead					
CEO Clarity (DWZ Qtr-Exp)	**-0.0083*****	**-0.0148*****	**-0.0093*****	**-0.0129*****	**-0.0092*****	**-0.0126*****	-0.0024	**-0.0215****	-0.0041	**-0.0210****	-0.0042	**-0.0210****
	(0.0033)	(0.0047)	(0.0033)	(0.0045)	(0.0033)	(0.0045)	(0.0053)	(0.0097)	(0.0053)	(0.0094)	(0.0053)	(0.0095)
CEO Residual Unc. (DWZ Qtr-Exp)	**0.0018***	**0.0020****	**0.0016***	**0.0018***	**0.0016***	**0.0017***	**0.0046*****	**0.0036****	**0.0044****	**0.0033****	**0.0044****	**0.0033****
	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0019)	(0.0017)	(0.0019)	(0.0017)	(0.0019)	(0.0017)
CEO Pres Uncertainty	0.0001	0.0006	-0.0001	0.0000	-0.0001	0.0001	0.0002	0.0016	-0.0003	0.0011	-0.0004	0.0011
	(0.0013)	(0.0013)	(0.0013)	(0.0013)	(0.0013)	(0.0013)	(0.0024)	(0.0021)	(0.0024)	(0.0021)	(0.0024)	(0.0021)
Leverage	**-0.0131*****	**-0.0324*****	**-0.0089*****	**-0.0199****	**-0.0087*****	**-0.0190****	**-0.0293****	**-0.0295****	**-0.0241****	-0.0181	**-0.0244****	-0.0179
	(0.0040)	(0.0086)	(0.0031)	(0.0085)	(0.0031)	(0.0085)	(0.0122)	(0.0139)	(0.0108)	(0.0136)	(0.0109)	(0.0136)
lnAssets	-0.0009	**-0.0169*****	**-0.0011***	**-0.0169*****	**-0.0011***	**-0.0167*****	**-0.0018***	**-0.0285*****	-0.0016	**-0.0286*****	-0.0015	**-0.0287*****
	(0.0006)	(0.0028)	(0.0006)	(0.0027)	(0.0006)	(0.0027)	(0.0011)	(0.0049)	(0.0010)	(0.0049)	(0.0011)	(0.0050)
TobinsQ	**0.0052*****	**0.0039*****	**0.0034*****	**0.0028****	**0.0033*****	**0.0025****	**0.0079*****	0.0011	**0.0046*****	-0.0009	**0.0046*****	-0.0009
	(0.0007)	(0.0011)	(0.0009)	(0.0012)	(0.0009)	(0.0012)	(0.0012)	(0.0017)	(0.0014)	(0.0017)	(0.0014)	(0.0018)
ROA	-0.0107	**0.0230***	**-0.0510*****	-0.0135	**-0.0498*****	-0.0129	**-0.0621*****	-0.0188	**-0.1280*****	**-0.0721*****	**-0.1270*****	**-0.0731*****
	(0.0109)	(0.0124)	(0.0131)	(0.0144)	(0.0132)	(0.0144)	(0.0192)	(0.0227)	(0.0202)	(0.0237)	(0.0204)	(0.0240)
Capex	**-0.0967*****	**-0.2472*****	**-0.1415*****	**-0.2628*****	**-0.1414*****	**-0.2605*****	**-0.1613*****	**-0.2816*****	**-0.2420*****	**-0.3002*****	**-0.2423*****	**-0.2996*****
	(0.0150)	(0.0253)	(0.0169)	(0.0258)	(0.0168)	(0.0257)	(0.0214)	(0.0372)	(0.0249)	(0.0375)	(0.0249)	(0.0375)
DivDummy	**-0.0037****	-0.0021	**-0.0051*****	-0.0034	**-0.0050*****	-0.0033	**-0.0042***	-0.0080	**-0.0048****	**-0.0090***	**-0.0047***	**-0.0091***
	(0.0015)	(0.0026)	(0.0015)	(0.0025)	(0.0015)	(0.0025)	(0.0025)	(0.0054)	(0.0025)	(0.0053)	(0.0025)	(0.0053)
sCFO	**0.0090*****	-0.0008	**0.0100*****	0.0004	**0.0099*****	0.0004	**0.0147***	0.0031	**0.0143***	0.0038	**0.0142***	0.0037
	(0.0024)	(0.0017)	(0.0031)	(0.0014)	(0.0030)	(0.0015)	(0.0081)	(0.0032)	(0.0082)	(0.0032)	(0.0081)	(0.0032)
Lagged_DV	**0.8718*****	**0.6433*****	**0.8734*****	**0.6576*****	**0.8738*****	**0.6579*****	**0.7424*****	**0.2158*****	**0.7404*****	**0.2320*****	**0.7402*****	**0.2317*****
	(0.0091)	(0.0159)	(0.0091)	(0.0156)	(0.0091)	(0.0156)	(0.0150)	(0.0242)	(0.0148)	(0.0242)	(0.0148)	(0.0242)
SalesGrowth			**-0.0165*****	**-0.0257*****	**-0.0165*****	**-0.0256*****			-0.0106	**-0.0137****	-0.0107	**-0.0136****
			(0.0054)	(0.0068)	(0.0054)	(0.0068)			(0.0070)	(0.0064)	(0.0070)	(0.0064)
RDSales			**0.0091*****	-0.0004	**0.0091*****	-0.0003			**0.0160*****	0.0005	**0.0159*****	0.0005
			(0.0025)	(0.0017)	(0.0025)	(0.0017)			(0.0038)	(0.0021)	(0.0038)	(0.0021)
CashFlowAt			**0.1182*****	**0.1536*****	**0.1189*****	**0.1544*****			**0.1977*****	**0.1883*****	**0.1974*****	**0.1880*****
			(0.0159)	(0.0175)	(0.0159)	(0.0175)			(0.0275)	(0.0300)	(0.0275)	(0.0300)
DailyVola			0.0000	**-0.0001****	0.0000	**-0.0001****			**0.0002*****	0.0000	**0.0003*****	0.0000
			(0.0000)	(0.0000)	(0.0000)	(0.0000)			(0.0001)	(0.0000)	(0.0001)	(0.0001)
Extended Controls			Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes
Industry FE	Yes		Yes		Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes		Yes		Yes
Year FE	Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes		
Year-Quarter FE					Yes	Yes					Yes	Yes
N	24,868	24,868	24,855	24,855	24,855	24,855	23,127	23,127	23,120	23,120	23,120	23,120
R^2	0.839	0.457	0.842	0.473	0.842	0.473	0.675	0.087	0.683	0.105	0.683	0.105
Adj. R^2	0.839	0.430	0.841	0.446	0.842	0.446	0.674	0.041	0.682	0.059	0.682	0.058

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). R^2 includes absorbed fixed effects (not within-R^2).

Table 4: H1.2 CEO 3-IV Decomp HFC: ClarityCEO + UncResCEO (DWZ Full) + UncPreCEO × Unrated Constraint

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	CashRatio				CashRatio_lead			
CEO Clarity (DWZ, c)	**-0.0062****	-0.0043	**-0.0059****	-0.0042	-0.0052	**-0.0157***	-0.0052	**-0.0155***
	(0.0034)	(0.0052)	(0.0034)	(0.0052)	(0.0053)	(0.0096)	(0.0053)	(0.0096)
CEO Residual Unc. (DWZ, c)	**0.0020****	**0.0018****	**0.0022****	**0.0019****	**0.0027****	**0.0022***	**0.0026****	**0.0021***
	(0.0011)	(0.0010)	(0.0011)	(0.0010)	(0.0015)	(0.0013)	(0.0015)	(0.0013)
CEO Pres Unc. (c)	0.0002	0.0005	0.0003	0.0007	-0.0007	-0.0001	-0.0008	-0.0002
	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0022)	(0.0018)	(0.0022)	(0.0018)
Unrated	**0.0038*****	-0.0027	-0.0010	-0.0026	**0.0065*****	**-0.0093***	-0.0014	**-0.0095***
	(0.0012)	(0.0035)	(0.0016)	(0.0035)	(0.0022)	(0.0054)	(0.0029)	(0.0054)
UncRes × Unrated	0.0011	-0.0001	0.0012	-0.0000	**0.0057****	0.0033	**0.0057****	0.0033
	(0.0021)	(0.0020)	(0.0021)	(0.0020)	(0.0029)	(0.0026)	(0.0029)	(0.0026)
UncPre × Unrated	0.0024	0.0013	**0.0029***	0.0014	0.0005	-0.0026	0.0010	-0.0027
	(0.0022)	(0.0025)	(0.0023)	(0.0025)	(0.0040)	(0.0038)	(0.0040)	(0.0038)
Leverage	**-0.0117*****	**-0.0216****	**-0.0141*****	**-0.0214****	**-0.0237****	-0.0107	**-0.0274****	-0.0104
	(0.0033)	(0.0098)	(0.0041)	(0.0099)	(0.0103)	(0.0131)	(0.0117)	(0.0131)
lnAssets	**0.0005***	**-0.0159*****	**-0.0019*****	**-0.0155*****	**0.0018*****	**-0.0273*****	**-0.0025****	**-0.0275*****
	(0.0003)	(0.0027)	(0.0007)	(0.0027)	(0.0005)	(0.0044)	(0.0011)	(0.0044)
TobinsQ	**0.0047*****	**0.0045*****	**0.0041*****	**0.0040*****	**0.0059*****	0.0009	**0.0052*****	0.0010
	(0.0008)	(0.0011)	(0.0008)	(0.0011)	(0.0013)	(0.0017)	(0.0013)	(0.0017)
ROA	**-0.0715*****	-0.0157	**-0.0651*****	-0.0133	**-0.1452*****	**-0.0819*****	**-0.1378*****	**-0.0800*****
	(0.0122)	(0.0122)	(0.0122)	(0.0122)	(0.0182)	(0.0200)	(0.0183)	(0.0202)
Capex	**-0.1610*****	**-0.3087*****	**-0.1647*****	**-0.3005*****	**-0.2338*****	**-0.3399*****	**-0.2426*****	**-0.3419*****
	(0.0158)	(0.0254)	(0.0158)	(0.0251)	(0.0226)	(0.0360)	(0.0227)	(0.0361)
DivDummy	**-0.0058*****	**-0.0043***	**-0.0054*****	-0.0038	**-0.0054****	-0.0065	**-0.0055****	-0.0066
	(0.0013)	(0.0023)	(0.0013)	(0.0023)	(0.0023)	(0.0044)	(0.0023)	(0.0044)
sCFO	**0.0081*****	-0.0003	**0.0072*****	-0.0005	**0.0135****	0.0043	**0.0125****	0.0042
	(0.0031)	(0.0016)	(0.0026)	(0.0016)	(0.0062)	(0.0029)	(0.0055)	(0.0029)
SalesGrowth	**-0.0189*****	**-0.0297*****	**-0.0176*****	**-0.0287*****	**-0.0158****	**-0.0156*****	**-0.0143****	**-0.0148*****
	(0.0045)	(0.0048)	(0.0044)	(0.0048)	(0.0062)	(0.0046)	(0.0062)	(0.0047)
RDSales	**0.0089*****	-0.0018	**0.0090*****	-0.0016	**0.0152*****	-0.0004	**0.0155*****	-0.0003
	(0.0020)	(0.0015)	(0.0020)	(0.0015)	(0.0032)	(0.0024)	(0.0032)	(0.0024)
CashFlowAt	**0.1415*****	**0.1741*****	**0.1368*****	**0.1729*****	**0.2109*****	**0.1796*****	**0.2012*****	**0.1767*****
	(0.0166)	(0.0180)	(0.0165)	(0.0180)	(0.0240)	(0.0257)	(0.0241)	(0.0259)
DailyVola	**0.0001****	**-0.0001*****	**0.0001****	-0.0001	**0.0004*****	**0.0001*****	**0.0003*****	0.0001
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0000)	(0.0001)	(0.0001)
Lagged_DV	**0.8605*****	**0.6188*****	**0.8571*****	**0.6204*****	**0.7310*****	**0.2231*****	**0.7250*****	**0.2242*****
	(0.0079)	(0.0133)	(0.0080)	(0.0134)	(0.0135)	(0.0193)	(0.0138)	(0.0193)
Industry FE	Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes
Year FE	Yes	Yes			Yes	Yes		
Year-Quarter FE			Yes	Yes			Yes	Yes
N	38,737	38,737	38,737	38,737	38,092	38,092	38,092	38,092
R^2	0.827	0.432	0.829	0.432	0.667	0.100	0.669	0.100
Adj. R^2	0.827	0.412	0.828	0.411	0.667	0.069	0.668	0.067

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Fiscal years 2002-2016. DWZ Eq.5 decomposition: Clarity = -CEO FE; UncRes = call-level residual. R^2 includes absorbed fixed effects (not within-R^2).

Table 5: H1.2 CEO 3-IV Decomp HFC: ClarityCEO + UncResCEO (DWZ Qtr-Exp, no-look-ahead) + UncPreCEO × Unrated Constraint

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	CashRatio				CashRatio_lead			
CEO Clarity (DWZ Qtr-Exp, c)	**-0.0114*****	**-0.0132*****	**-0.0104*****	**-0.0127*****	-0.0037	**-0.0186****	-0.0019	**-0.0187****
	(0.0035)	(0.0052)	(0.0034)	(0.0053)	(0.0054)	(0.0103)	(0.0054)	(0.0103)
CEO Residual Unc. (DWZ Qtr-Exp, c)	**0.0021***	**0.0024****	**0.0021***	**0.0022****	**0.0044****	**0.0036****	**0.0044****	**0.0036****
	(0.0013)	(0.0013)	(0.0013)	(0.0013)	(0.0020)	(0.0017)	(0.0020)	(0.0017)
CEO Pres Unc. (c)	0.0000	-0.0005	0.0000	-0.0004	0.0003	0.0009	0.0000	0.0009
	(0.0013)	(0.0013)	(0.0013)	(0.0013)	(0.0025)	(0.0022)	(0.0025)	(0.0022)
Unrated	0.0021	-0.0043	-0.0012	-0.0043	**0.0051****	-0.0073	-0.0019	-0.0074
	(0.0014)	(0.0043)	(0.0018)	(0.0043)	(0.0024)	(0.0066)	(0.0032)	(0.0066)
UncRes (Qtr-Exp) × Unrated	0.0019	0.0013	0.0019	0.0013	**0.0050***	0.0023	**0.0050***	0.0022
	(0.0025)	(0.0023)	(0.0025)	(0.0023)	(0.0038)	(0.0033)	(0.0038)	(0.0033)
UncPre × Unrated	0.0014	-0.0014	0.0017	-0.0014	-0.0001	-0.0015	0.0004	-0.0014
	(0.0026)	(0.0029)	(0.0026)	(0.0029)	(0.0046)	(0.0046)	(0.0046)	(0.0046)
Leverage	**-0.0084*****	-0.0170	**-0.0094*****	-0.0160	**-0.0210***	-0.0110	**-0.0240****	-0.0107
	(0.0031)	(0.0104)	(0.0034)	(0.0104)	(0.0108)	(0.0154)	(0.0121)	(0.0154)
lnAssets	0.0003	**-0.0157*****	**-0.0016***	**-0.0157*****	**0.0016*****	**-0.0300*****	**-0.0021***	**-0.0302*****
	(0.0003)	(0.0031)	(0.0008)	(0.0031)	(0.0006)	(0.0052)	(0.0013)	(0.0053)
TobinsQ	**0.0039*****	**0.0030****	**0.0034*****	**0.0025***	**0.0049*****	-0.0011	**0.0044*****	-0.0011
	(0.0010)	(0.0015)	(0.0010)	(0.0014)	(0.0014)	(0.0019)	(0.0015)	(0.0020)
ROA	**-0.0554*****	-0.0071	**-0.0535*****	-0.0065	**-0.1336*****	**-0.0761*****	**-0.1305*****	**-0.0773*****
	(0.0139)	(0.0145)	(0.0140)	(0.0146)	(0.0211)	(0.0249)	(0.0213)	(0.0252)
Capex	**-0.1396*****	**-0.2564*****	**-0.1439*****	**-0.2536*****	**-0.2276*****	**-0.2870*****	**-0.2378*****	**-0.2865*****
	(0.0175)	(0.0276)	(0.0176)	(0.0275)	(0.0263)	(0.0384)	(0.0262)	(0.0384)
DivDummy	**-0.0055*****	-0.0041	**-0.0056*****	-0.0041	**-0.0044***	-0.0076	**-0.0046***	-0.0077
	(0.0016)	(0.0027)	(0.0016)	(0.0026)	(0.0025)	(0.0053)	(0.0026)	(0.0053)
sCFO	**0.0099*****	0.0017	**0.0094*****	0.0017	**0.0123***	0.0041	**0.0116***	0.0040
	(0.0030)	(0.0016)	(0.0028)	(0.0016)	(0.0069)	(0.0032)	(0.0063)	(0.0032)
SalesGrowth	**-0.0128****	**-0.0267*****	**-0.0128****	**-0.0265*****	-0.0081	**-0.0127****	-0.0083	**-0.0127****
	(0.0051)	(0.0056)	(0.0051)	(0.0056)	(0.0074)	(0.0063)	(0.0075)	(0.0063)
RDSales	**0.0086*****	-0.0004	**0.0086*****	-0.0003	**0.0154*****	0.0010	**0.0154*****	0.0011
	(0.0022)	(0.0017)	(0.0022)	(0.0017)	(0.0035)	(0.0022)	(0.0035)	(0.0022)
CashFlowAt	**0.1220*****	**0.1745*****	**0.1204*****	**0.1754*****	**0.2021*****	**0.1992*****	**0.1968*****	**0.1990*****
	(0.0169)	(0.0188)	(0.0169)	(0.0189)	(0.0288)	(0.0317)	(0.0288)	(0.0317)
DailyVola	0.0000	**-0.0001****	-0.0000	**-0.0001****	**0.0003*****	0.0000	**0.0002*****	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0000)	(0.0001)	(0.0001)
Lagged_DV	**0.8737*****	**0.6448*****	**0.8713*****	**0.6451*****	**0.7471*****	**0.2124*****	**0.7408*****	**0.2122*****
	(0.0096)	(0.0166)	(0.0096)	(0.0166)	(0.0152)	(0.0230)	(0.0154)	(0.0229)
Industry FE	Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes
Year FE	Yes	Yes			Yes	Yes		
Year-Quarter FE			Yes	Yes			Yes	Yes
N	21,824	21,824	21,824	21,824	21,371	21,371	21,371	21,371
R^2	0.841	0.453	0.841	0.453	0.679	0.094	0.680	0.094
Adj. R^2	0.841	0.423	0.841	0.423	0.678	0.046	0.679	0.044

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Fiscal years 2002-2016. DWZ Eq.5 decomposition under quarterly-expanding window (no-look-ahead). R^2 includes absorbed fixed effects (not within-R^2).

Table 6: H1.3 CEO 3-IV Decomp CFvol: ClarityCEO + UncResCEO (DWZ Full) + UncPreCEO × HighCFvol (Han-Qiu 16q CV)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	CashRatio				CashRatio_lead			
CEO Clarity (DWZ, c)	**-0.0088*****	-0.0059	**-0.0071****	-0.0058	**-0.0096****	**-0.0135***	**-0.0078***	**-0.0134***
	(0.0034)	(0.0047)	(0.0033)	(0.0047)	(0.0052)	(0.0091)	(0.0052)	(0.0091)
CEO Residual Unc. (DWZ, c)	**0.0021****	**0.0017***	**0.0022****	**0.0017****	**0.0025****	0.0016	**0.0024***	0.0015
	(0.0011)	(0.0010)	(0.0011)	(0.0010)	(0.0015)	(0.0013)	(0.0015)	(0.0014)
CEO Pres Unc. (c)	0.0004	0.0006	0.0001	0.0007	-0.0002	0.0004	-0.0008	0.0002
	(0.0011)	(0.0011)	(0.0012)	(0.0012)	(0.0021)	(0.0018)	(0.0022)	(0.0018)
HighCFvol	-0.0005	-0.0022	**-0.0028****	-0.0023	**0.0075*****	0.0021	**0.0049****	0.0021
	(0.0012)	(0.0016)	(0.0013)	(0.0016)	(0.0019)	(0.0025)	(0.0020)	(0.0025)
UncRes × HighCFvol	0.0004	0.0004	-0.0000	0.0002	**0.0047***	0.0028	**0.0048***	0.0030
	(0.0020)	(0.0019)	(0.0020)	(0.0019)	(0.0030)	(0.0026)	(0.0029)	(0.0026)
UncPre × HighCFvol	**0.0048****	0.0002	**0.0049****	0.0002	**0.0099*****	0.0040	**0.0101*****	0.0041
	(0.0022)	(0.0024)	(0.0022)	(0.0024)	(0.0039)	(0.0040)	(0.0039)	(0.0040)
Leverage	**-0.0127*****	**-0.0191****	**-0.0130*****	**-0.0189****	**-0.0273*****	-0.0116	**-0.0274*****	-0.0112
	(0.0034)	(0.0082)	(0.0036)	(0.0082)	(0.0094)	(0.0121)	(0.0097)	(0.0121)
lnAssets	0.0004	**-0.0174*****	**-0.0015*****	**-0.0169*****	**0.0014*****	**-0.0236*****	-0.0013	**-0.0235*****
	(0.0003)	(0.0026)	(0.0006)	(0.0026)	(0.0005)	(0.0043)	(0.0010)	(0.0044)
TobinsQ	**0.0044*****	**0.0037*****	**0.0035*****	**0.0034*****	**0.0066*****	0.0016	**0.0056*****	0.0017
	(0.0008)	(0.0011)	(0.0008)	(0.0011)	(0.0013)	(0.0017)	(0.0014)	(0.0018)
ROA	**-0.0641*****	-0.0148	**-0.0563*****	-0.0121	**-0.1405*****	**-0.0736*****	**-0.1330*****	**-0.0717*****
	(0.0117)	(0.0123)	(0.0116)	(0.0123)	(0.0185)	(0.0205)	(0.0186)	(0.0206)
Capex	**-0.1692*****	**-0.3164*****	**-0.1779*****	**-0.3085*****	**-0.2443*****	**-0.3482*****	**-0.2565*****	**-0.3496*****
	(0.0158)	(0.0251)	(0.0160)	(0.0249)	(0.0229)	(0.0363)	(0.0232)	(0.0364)
DivDummy	**-0.0057*****	**-0.0051****	**-0.0054*****	**-0.0046****	**-0.0048****	**-0.0085***	**-0.0049****	**-0.0086***
	(0.0013)	(0.0023)	(0.0013)	(0.0023)	(0.0023)	(0.0046)	(0.0023)	(0.0046)
sCFO	**0.0087****	0.0004	**0.0078****	0.0003	**0.0128***	0.0041	**0.0121***	0.0040
	(0.0043)	(0.0017)	(0.0037)	(0.0017)	(0.0071)	(0.0031)	(0.0066)	(0.0030)
SalesGrowth	**-0.0203*****	**-0.0291*****	**-0.0192*****	**-0.0283*****	**-0.0187*****	**-0.0188*****	**-0.0172****	**-0.0181*****
	(0.0050)	(0.0057)	(0.0049)	(0.0057)	(0.0068)	(0.0055)	(0.0067)	(0.0056)
RDSales	**0.0063*****	**-0.0031****	**0.0063*****	**-0.0029****	**0.0153*****	-0.0015	**0.0154*****	-0.0014
	(0.0022)	(0.0015)	(0.0022)	(0.0015)	(0.0037)	(0.0029)	(0.0036)	(0.0029)
CashFlowAt	**0.1476*****	**0.1713*****	**0.1382*****	**0.1698*****	**0.2228*****	**0.1725*****	**0.2093*****	**0.1698*****
	(0.0170)	(0.0179)	(0.0171)	(0.0179)	(0.0255)	(0.0262)	(0.0258)	(0.0264)
DailyVola	**0.0001*****	**-0.0001*****	**0.0001*****	-0.0001	**0.0004*****	**0.0001*****	**0.0003*****	0.0001
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0000)	(0.0001)	(0.0001)
Lagged_DV	**0.8664*****	**0.6307*****	**0.8619*****	**0.6322*****	**0.7262*****	**0.2312*****	**0.7200*****	**0.2320*****
	(0.0077)	(0.0133)	(0.0078)	(0.0133)	(0.0139)	(0.0205)	(0.0142)	(0.0205)
Industry FE	Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes
Year FE	Yes	Yes			Yes	Yes		
Year-Quarter FE			Yes	Yes			Yes	Yes
N	40,795	40,795	40,795	40,795	38,671	38,671	38,671	38,671
R^2	0.825	0.447	0.826	0.447	0.657	0.103	0.659	0.102
Adj. R^2	0.824	0.429	0.826	0.428	0.657	0.072	0.658	0.071

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Fiscal years 2002-2018. Full panel coverage via Compustat Quarterly OCF Extended (1990-2002 backfill); Han-Qiu 16-trailing-quarter CFvol available from 1994-Q1 onwards in source data. R^2 includes absorbed fixed effects (not within-R^2).

Table 7: H11: Political Risk and Language Uncertainty

	(1)	(2)	(3)	(4)
	Industry FE		Firm FE	
	UncResCEO	UncPreCEO	UncResCEO	UncPreCEO
Political Risk$_t$	**0.0001*****	**0.0003*****	**0.0001*****	**0.0002*****
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
UncQue	-0.0022	**0.0204*****	-0.0014	**0.0081*****
	(0.0032)	(0.0045)	(0.0037)	(0.0029)
NegCall	-0.0025	**0.1874*****	-0.0036	**0.1477*****
	(0.0051)	(0.0141)	(0.0073)	(0.0082)
lnAssets	-0.0002	**-0.0296*****	-0.0054	0.0048
	(0.0004)	(0.0039)	(0.0058)	(0.0078)
TobinsQ	-0.0005	-0.0007	-0.0008	0.0001
	(0.0009)	(0.0036)	(0.0022)	(0.0023)
ROA	**0.0291****	0.0477	**0.0540****	0.0346
	(0.0145)	(0.0318)	(0.0264)	(0.0230)
CashRatio	0.0019	-0.0354	0.0206	0.0265
	(0.0077)	(0.0357)	(0.0240)	(0.0287)
DivDummy	-0.0031	-0.0139	-0.0108	**-0.0229****
	(0.0021)	(0.0118)	(0.0074)	(0.0103)
FirmMat	0.0012	**0.0021****	0.0027	0.0006
	(0.0013)	(0.0010)	(0.0041)	(0.0007)
EarnVol	0.0078	-0.0072	0.0056	-0.0104
	(0.0077)	(0.0157)	(0.0117)	(0.0132)
UncPreCEO	-0.0051		-0.0054	
	(0.0034)		(0.0054)	
Industry FE	Yes	Yes		
Firm FE			Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	44,497	65,760	44,497	65,760
R^2	0.003	0.043	0.003	0.023
Adj. R^2	0.002	0.043	-0.030	-0.004

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Firms with fewer than 5 calls are excluded. R^2 includes absorbed fixed effects (not within-R^2).

Table 8: H11-Lag2: Political Risk (2-Qtr Lag) and Language Uncertainty

	(1)	(2)	(3)	(4)
	Industry FE		Firm FE	
	UncResCEO	UncPreCEO	UncResCEO	UncPreCEO
PRisk_{t-2}	0.0000	**0.0001*****	0.0000	**0.0000*****
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
UncQue	0.0001	**0.0242*****	0.0010	**0.0099*****
	(0.0033)	(0.0047)	(0.0038)	(0.0031)
NegCall	0.0020	**0.2019*****	0.0019	**0.1609*****
	(0.0052)	(0.0145)	(0.0074)	(0.0084)
lnAssets	-0.0001	**-0.0298*****	-0.0035	0.0047
	(0.0005)	(0.0040)	(0.0062)	(0.0078)
TobinsQ	-0.0005	-0.0002	-0.0011	-0.0002
	(0.0010)	(0.0037)	(0.0023)	(0.0023)
ROA	**0.0293***	0.0518	**0.0598****	**0.0464***
	(0.0152)	(0.0331)	(0.0272)	(0.0247)
CashRatio	0.0064	-0.0400	0.0170	0.0233
	(0.0076)	(0.0368)	(0.0242)	(0.0292)
DivDummy	-0.0031	-0.0142	**-0.0133***	**-0.0207***
	(0.0022)	(0.0121)	(0.0076)	(0.0106)
FirmMat	0.0013	0.0021	0.0025	-0.0007
	(0.0013)	(0.0015)	(0.0041)	(0.0016)
EarnVol	0.0096	-0.0078	0.0089	-0.0154
	(0.0063)	(0.0159)	(0.0100)	(0.0158)
UncPreCEO	-0.0001		-0.0003	
	(0.0035)		(0.0056)	
Industry FE	Yes	Yes		
Firm FE			Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	43,037	63,022	43,037	63,022
R^2	0.000	0.035	0.000	0.015
Adj. R^2	-0.001	0.034	-0.033	-0.013

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Firms with fewer than 5 calls are excluded. R^2 includes absorbed fixed effects (not within-R^2).

Table 9: H23: Product-Market Competition and Uncertainty Language

	(1)	(2)	(3)	(4)
	Industry FE		Firm FE	
	UncResCEO	UncPreCEO	UncResCEO	UncPreCEO
$z(\log(\text{TSIMM}))$	0.0008	**0.0304******	0.0023	**0.0302******
	(0.0014)	(0.0074)	(0.0075)	(0.0093)
UncQue	0.0011	**0.0398******	0.0039	0.0084
	(0.0062)	(0.0100)	(0.0087)	(0.0073)
NegCall	0.0020	**0.2340******	0.0046	**0.1698******
	(0.0068)	(0.0201)	(0.0120)	(0.0140)
lnAssets	-0.0007	**-0.0335******	-0.0101	0.0013
	(0.0007)	(0.0038)	(0.0071)	(0.0081)
TobinsQ	0.0001	-0.0007	0.0009	-0.0017
	(0.0013)	(0.0037)	(0.0025)	(0.0025)
ROA	0.0233	**0.0681*****	0.0343	0.0241
	(0.0197)	(0.0314)	(0.0308)	(0.0254)
CashRatio	0.0065	**-0.0803*****	0.0055	0.0022
	(0.0103)	(0.0360)	(0.0287)	(0.0306)
DivDummy	-0.0020	-0.0060	-0.0074	-0.0166
	(0.0028)	(0.0117)	(0.0083)	(0.0106)
FirmMat	0.0018	**0.0014*****	0.0083	0.0003
	(0.0019)	(0.0007)	(0.0056)	(0.0004)
EarnVol	-0.0047	-0.0210	-0.0048	-0.0114
	(0.0087)	(0.0152)	(0.0121)	(0.0146)
UncPreCEO	**0.0111*****		**0.0230*****	
	(0.0044)		(0.0093)	
Industry FE	Yes	Yes		
Firm FE			Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	12,728	18,492	12,728	18,492
R^2	0.001	0.050	0.001	0.021
Adj. R^2	-0.002	0.048	-0.098	-0.068

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Unit of observation: firm-fiscal-year. R^2 includes absorbed fixed effects (not within-R^2).

Table 10: H24: US Economic Policy Uncertainty and Call Language Uncertainty

	(1)	(2)	(3)	(4)
	Industry + Cal. Year FE		Firm + Cal. Year FE	
	UncResCEO	UncPreCEO	UncResCEO	UncPreCEO
$\log(\text{US EPU})_t$	**0.0124****	**0.0211*****	**0.0123***	**0.0239*****
	(0.0075)	(0.0074)	(0.0075)	(0.0076)
UncQue	0.0011	**0.0164*****	0.0024	**0.0095*****
	(0.0035)	(0.0032)	(0.0040)	(0.0028)
NegCall	0.0006	**0.1247*****	0.0021	**0.1385*****
	(0.0056)	(0.0089)	(0.0073)	(0.0079)
lnAssets	0.0009	**-0.0143*****	0.0013	0.0038
	(0.0007)	(0.0020)	(0.0067)	(0.0063)
TobinsQ	-0.0001	0.0007	-0.0000	0.0007
	(0.0011)	(0.0019)	(0.0027)	(0.0018)
ROA	**0.0281***	**0.0367****	**0.0677****	**0.0467****
	(0.0149)	(0.0168)	(0.0268)	(0.0191)
CashRatio	0.0051	-0.0079	0.0345	0.0305
	(0.0087)	(0.0187)	(0.0256)	(0.0231)
DivDummy	**-0.0046*****	-0.0074	**-0.0142****	**-0.0157***
	(0.0009)	(0.0061)	(0.0072)	(0.0083)
FirmMat	**0.0023***	0.0009	0.0005	-0.0011
	(0.0012)	(0.0008)	(0.0047)	(0.0012)
EarnVol	**0.0268***	-0.0074	**0.0353****	-0.0083
	(0.0157)	(0.0086)	(0.0176)	(0.0119)
UncPreCEO	-0.0026		-0.0011	
	(0.0032)		(0.0045)	
Lagged_DV	**0.0202*****	**0.5176*****	**0.0187*****	**0.2626*****
	(0.0054)	(0.0131)	(0.0056)	(0.0089)
Industry FE	Yes	Yes		
Firm FE			Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	39,063	60,503	39,063	60,503
R^2	0.001	0.292	0.001	0.084
Adj. R^2	-0.000	0.291	-0.035	0.057

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) two-way clustered (firm, calendar quarter). Main sample (excludes financial and utility firms). Firms with fewer than 5 calls are excluded. R^2 includes absorbed fixed effects (not within-R^2).

Table 11: H24b: Global Economic Policy Uncertainty and Call Language Uncertainty

	(1)	(2)	(3)	(4)
	Industry + Cal. Year FE		Firm + Cal. Year FE	
	UncResCEO	UncPreCEO	UncResCEO	UncPreCEO
$\log(\text{GEPU})_t$	**0.0181****	**0.0271*****	**0.0187****	**0.0309*****
	(0.0100)	(0.0107)	(0.0101)	(0.0108)
UncQue	0.0011	**0.0164*****	0.0024	**0.0095*****
	(0.0035)	(0.0032)	(0.0039)	(0.0028)
NegCall	0.0006	**0.1247*****	0.0020	**0.1385*****
	(0.0056)	(0.0089)	(0.0072)	(0.0079)
lnAssets	0.0009	**-0.0143*****	0.0012	0.0037
	(0.0007)	(0.0020)	(0.0067)	(0.0063)
TobinsQ	-0.0001	0.0008	0.0000	0.0007
	(0.0011)	(0.0019)	(0.0027)	(0.0018)
ROA	**0.0280***	**0.0367****	**0.0673****	**0.0462****
	(0.0149)	(0.0168)	(0.0267)	(0.0191)
CashRatio	0.0051	-0.0080	0.0345	0.0304
	(0.0087)	(0.0188)	(0.0256)	(0.0231)
DivDummy	**-0.0047*****	-0.0074	**-0.0142****	**-0.0156***
	(0.0009)	(0.0061)	(0.0072)	(0.0083)
FirmMat	**0.0023***	0.0009	0.0006	-0.0011
	(0.0012)	(0.0008)	(0.0047)	(0.0012)
EarnVol	**0.0269***	-0.0074	**0.0354****	-0.0082
	(0.0157)	(0.0086)	(0.0176)	(0.0119)
UncPreCEO	-0.0026		-0.0011	
	(0.0032)		(0.0045)	
Lagged_DV	**0.0202*****	**0.5176*****	**0.0187*****	**0.2626*****
	(0.0054)	(0.0131)	(0.0056)	(0.0089)
Industry FE	Yes	Yes		
Firm FE			Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	39,063	60,503	39,063	60,503
R^2	0.001	0.292	0.001	0.084
Adj. R^2	-0.000	0.291	-0.035	0.057

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) two-way clustered (firm, calendar quarter). Main sample (excludes financial and utility firms). Firms with fewer than 5 calls are excluded. R^2 includes absorbed fixed effects (not within-R^2).

Table 12: H14c CEO 3-IV Decomp: ClarityCEO + UncResCEO (DWZ Eq.5 Full) + UncPreCEO (raw) and 25-Day Post-Call Bid-Ask Spread Level ($[0, +25]$, the call day plus 25 trading days)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Spread$_{25D,t}$						Spread$_{25D,t+1}$					
CEO Clarity (DWZ)	-0.0477	1.0617	0.2122	1.1056	0.1644	1.0547	-0.1886	0.5998	-0.0922	0.6828	-0.0431	0.5722
	(0.2145)	(0.4277)	(0.2227)	(0.4260)	(0.2103)	(0.4174)	(0.2370)	(0.4004)	(0.2399)	(0.4055)	(0.2206)	(0.3729)
CEO Residual Uncertainty (DWZ)	-0.0594	-0.0687	-0.0882	-0.1021	-0.0757	-0.0893	0.0825	0.0696	0.0742	0.0545	0.0984	0.0751
	(0.1068)	(0.1057)	(0.1023)	(0.1007)	(0.0994)	(0.0981)	(0.1416)	(0.1437)	(0.1407)	(0.1426)	(0.1339)	(0.1357)
CEO Pres Uncertainty	**0.1664****	**0.3496*****	0.0814	**0.2945*****	0.0108	**0.1644***	-0.0817	-0.0288	-0.1072	-0.0360	0.0170	0.1124
	(0.0946)	(0.1253)	(0.0956)	(0.1229)	(0.0919)	(0.1184)	(0.0926)	(0.1229)	(0.0926)	(0.1233)	(0.0872)	(0.1186)
lnAssets	**-0.8039*****	**-1.9098*****	**-0.6339*****	**-1.6825*****	**-0.6352*****	**-1.8277*****	**-0.8218*****	**-1.5191*****	**-0.7857*****	**-1.7292*****	**-0.6568*****	**-1.4181*****
	(0.0490)	(0.2000)	(0.0472)	(0.2156)	(0.0473)	(0.2158)	(0.0513)	(0.1951)	(0.0505)	(0.2130)	(0.0474)	(0.2017)
TobinsQ	**-0.2861*****	**-0.4785*****	**-0.2633*****	**-0.5518*****	**-0.2654*****	**-0.5883*****	**-0.2305*****	**-0.3215*****	**-0.2285*****	**-0.4803*****	**-0.2035*****	**-0.4289*****
	(0.0457)	(0.0900)	(0.0470)	(0.0963)	(0.0451)	(0.0956)	(0.0467)	(0.0850)	(0.0478)	(0.1002)	(0.0433)	(0.0935)
ROA	**-9.4469*****	**-11.6430*****	**-5.7059*****	**-7.6844*****	**-5.7511*****	**-7.8517*****	**-9.9256*****	**-12.8443*****	**-8.8744*****	**-11.9197*****	**-7.6375*****	**-10.5867*****
	(1.0201)	(1.3867)	(0.9822)	(1.2654)	(0.9633)	(1.2490)	(1.0798)	(1.2992)	(1.0722)	(1.2732)	(0.9869)	(1.1921)
Leverage	**0.3875***	**1.8330*****	-0.1126	0.6850	-0.0542	0.8026	0.2924	0.5282	0.1338	0.2221	0.0592	0.1928
	(0.2088)	(0.5143)	(0.2434)	(0.5013)	(0.2264)	(0.4954)	(0.2224)	(0.5516)	(0.2248)	(0.5596)	(0.2111)	(0.5258)
Capex	-0.8001	0.9540	**-2.3622****	-1.5271	**-2.0790***	-1.2897	**-1.6463***	-3.2121	**-2.1561****	**-4.3379***	-1.4770	-2.4356
	(1.1109)	(2.9126)	(1.1044)	(2.8061)	(1.0845)	(2.7998)	(0.9620)	(2.2020)	(0.9823)	(2.2446)	(0.9011)	(2.1565)
DivDummy	0.1043	0.1429	**0.3329*****	0.2549	**0.3156*****	0.2723	0.0240	0.1007	0.1012	0.1454	0.1145	0.1739
	(0.0893)	(0.2020)	(0.0896)	(0.1888)	(0.0855)	(0.1867)	(0.0887)	(0.2095)	(0.0889)	(0.2074)	(0.0825)	(0.1975)
sCFO	**0.6700****	**0.8405****	0.4045	**0.6825***	0.4084	**0.6882****	0.0871	**0.3644***	0.0040	0.3165	-0.0840	0.2479
	(0.3405)	(0.3472)	(0.4604)	(0.3789)	(0.4126)	(0.3375)	(0.1759)	(0.1877)	(0.2174)	(0.2055)	(0.1937)	(0.1738)
Lagged_DV	**0.7547*****	**0.6569*****	**0.7140*****	**0.6180*****	**0.7310*****	**0.6318*****	**0.7540*****	**0.6361*****	**0.7360*****	**0.6135*****	**0.7676*****	**0.6518*****
	(0.0150)	(0.0186)	(0.0167)	(0.0199)	(0.0168)	(0.0208)	(0.0156)	(0.0185)	(0.0173)	(0.0205)	(0.0162)	(0.0197)
StockPrice			-0.0011	-0.0033	0.0002	0.0019			-0.0001	**0.0101*****	-0.0016	**0.0047****
			(0.0014)	(0.0024)	(0.0013)	(0.0024)			(0.0013)	(0.0024)	(0.0012)	(0.0022)
Turnover			**-24.1012*****	**-23.8647*****	**-20.7840*****	**-20.3411*****			**-7.9396*****	**-5.5050*****	**-7.1303*****	**-4.7073****
			(1.7508)	(2.2119)	(1.7290)	(2.1791)			(1.5411)	(2.0904)	(1.4699)	(1.9633)
DailyVola			**0.1229*****	**0.1393*****	**0.1075*****	**0.1257*****			**0.0404*****	**0.0520*****	**0.0396*****	**0.0441*****
			(0.0072)	(0.0081)	(0.0082)	(0.0099)			(0.0062)	(0.0068)	(0.0069)	(0.0081)
AbsSurpDec			**-0.0919*****	-0.0241	**-0.0756*****	-0.0093			-0.0050	0.0119	-0.0174	-0.0012
			(0.0265)	(0.0300)	(0.0253)	(0.0286)			(0.0278)	(0.0304)	(0.0261)	(0.0285)
Extended Controls			Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes
Industry FE	Yes		Yes		Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes		Yes		Yes
Year FE	Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes		
Year-Quarter FE					Yes	Yes					Yes	Yes
N	42,625	42,625	42,625	42,625	42,625	42,625	43,049	43,049	43,049	43,049	43,049	43,049
R^2	0.703	0.517	0.721	0.551	0.730	0.548	0.700	0.499	0.702	0.504	0.723	0.523
Adj. R^2	0.703	0.501	0.721	0.535	0.730	0.532	0.700	0.482	0.702	0.487	0.723	0.506

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). DWZ Full decomposition: Clarity = -CEO FE; UncRes = call-level residual. R^2 includes absorbed fixed effects (not within-R^2). Spread$_{25D}$ values rescaled by 10^4 (basis points) for readability; multiply coefficients by 10^{-4} to recover raw fraction units.

Table 13: H18 CEO 3-IV Decomp: ClarityCEO + UncResCEO (DWZ Eq.5 Full) + UncPreCEO (raw) and SEC Conference-Call Comment Letter Receipt (LPM, $[call_t, call_{t+1}]$ window)

	(1)	(2)	(3)	(4)	(5)	(6)
			CCCL			
CEO Clarity (DWZ)	0.0059	0.0039	0.0062	0.0040	0.0058	0.0040
	(0.0015)	(0.0030)	(0.0015)	(0.0030)	(0.0015)	(0.0030)
CEO Residual Uncertainty (DWZ)	-0.0003	-0.0003	-0.0003	-0.0003	-0.0004	-0.0004
	(0.0009)	(0.0009)	(0.0009)	(0.0009)	(0.0009)	(0.0009)
CEO Pres Uncertainty	0.0008	**0.0014***	**0.0016****	**0.0014***	0.0007	**0.0013***
	(0.0008)	(0.0009)	(0.0007)	(0.0009)	(0.0008)	(0.0009)
lnAssets	**-0.0006****	**0.0018***	0.0001	0.0017	**-0.0006****	**0.0019***
	(0.0002)	(0.0011)	(0.0001)	(0.0010)	(0.0003)	(0.0011)
TobinsQ	-0.0003	0.0002	-0.0001	0.0002	-0.0002	0.0004
	(0.0003)	(0.0004)	(0.0003)	(0.0004)	(0.0003)	(0.0004)
ROA	0.0006	0.0012	-0.0014	-0.0006	-0.0006	-0.0004
	(0.0034)	(0.0052)	(0.0052)	(0.0062)	(0.0051)	(0.0062)
Leverage	-0.0008	-0.0019	-0.0009	-0.0011	-0.0009	-0.0012
	(0.0015)	(0.0034)	(0.0015)	(0.0035)	(0.0015)	(0.0035)
Capex	-0.0029	0.0106	0.0004	0.0111	-0.0033	0.0107
	(0.0063)	(0.0104)	(0.0064)	(0.0104)	(0.0066)	(0.0103)
CashRatio	-0.0014	**-0.0075***	0.0009	**-0.0080***	-0.0014	**-0.0082***
	(0.0027)	(0.0044)	(0.0026)	(0.0044)	(0.0028)	(0.0044)
DivDummy	-0.0005	0.0003	-0.0005	0.0002	-0.0005	0.0002
	(0.0008)	(0.0015)	(0.0008)	(0.0015)	(0.0008)	(0.0015)
sCFO	-0.0007	-0.0020	-0.0004	-0.0019	-0.0007	-0.0019
	(0.0005)	(0.0024)	(0.0004)	(0.0024)	(0.0005)	(0.0024)
Lagged_DV	**-0.0079*****	**-0.0409*****	**-0.0077*****	**-0.0410*****	**-0.0074*****	**-0.0405*****
	(0.0008)	(0.0027)	(0.0007)	(0.0027)	(0.0008)	(0.0028)
SalesGrowth			-0.0002	-0.0013	-0.0001	-0.0011
			(0.0008)	(0.0011)	(0.0008)	(0.0011)
RDSales			-0.0003	-0.0002	-0.0003	-0.0001
			(0.0002)	(0.0002)	(0.0002)	(0.0002)
CashFlowAt			0.0025	0.0043	0.0009	0.0037
			(0.0050)	(0.0064)	(0.0050)	(0.0064)
DailyVola			0.0000	**-0.0000***	0.0000	-0.0000
			(0.0000)	(0.0000)	(0.0000)	(0.0000)
Extended Controls			Yes	Yes	Yes	Yes
Industry FE	Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes
Year FE	Yes	Yes	Yes	Yes		
Year-Quarter FE					Yes	Yes
N	44,113	44,113	44,096	44,096	44,096	44,096
R^2	0.001	0.002	0.000	0.002	0.001	0.002
Adj. R^2	-0.000	-0.032	-0.001	-0.032	-0.001	-0.033

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). DWZ Full decomposition: Clarity = -CEO FE; UncRes = call-level residual. R^2 includes absorbed fixed effects (not within-R^2).

A Variable Definitions

This appendix lists every variable used in the empirical analyses, with construction formula, data source, and originating reference. Variable names follow paper-standard conventions where available (Dzieliński et al. 2021 for speech measures, Bates et al. 2009 for cash-holdings controls); deviations from source-paper frequency or sample are documented in the formula column.

A.1 Sample Manifest

Name	Description / Formula	Source	Reference
`file_name`, `ceo_id`, `ceo_name`, `gvkey`, `ff12_code`, `ff12_name`, `start_date`	Master sample manifest with CEO, firm, and industry identifiers	outputs/1.4_AssembleManifest	

A.2 Text/Linguistic Variables

Name	Description / Formula	Source	Reference
`ClarityCEO`	CEO clarity persistent-style component (DWZ Eqn 5). *Formula:* $\text{ClarityCEO}_i = -\widehat{\text{FE}}_i$ where $\widehat{\text{FE}}_i$ is the CEO-i fixed-effect estimate from the panel regression $\text{UncAnsCEO}_{ic} = \alpha + \sum_i \text{FE}_i \cdot \mathbb{1}[\text{CEO} = i] + \mathbf{X}_{ic}\gamma + \varepsilon_{ic}$ over the call-level Q&A uncertainty panel; controls $\mathbf{X}$ are the DWZ extended-controls vector. The sign flip on $\widehat{\text{FE}}_i$ makes higher ClarityCEO mean "CEO speaks more clearly" (lower mean uncertainty). Persistent CEO trait.	outputs/econometric/ceo_clarity_extended/	Dzielínski et al. (2021) (Section 4.4 Eqn 5, p.16)
`ClarityCEO_QtrExp`	ClarityCEO under within-tenure expanding window. *Formula:* As ClarityCEO, but the CEO fixed-effect regression is re-estimated on an expanding within-CEO window using only calls of CEO i available up to and including focal call c. Strictly look-ahead-free.	outputs/econometric/ceo_clarity_expanding/	thesis QtrExp variant of Dzielínski et al. (2021) Eqn 5
`NegCall`	Entire-call negative sentiment percentage. *Formula:* Ratio of negative words to total words in the entire call, based on LM 2011 negative-words list. Per DWZ 2021 Appendix Table A.1: 'percentage of negative words in all words spoken by the CEO, CFO and analysts attending the call.'	outputs/2_Textual_Analysis/2.2_Variables	Dzielínski et al. (2021) (Section 4.4, p.18; Appendix Table A.1, p.54)
`UncAnsCEO`	CEO-only Q&A uncertainty percentage (input to ClarityCEO/UncResCEO residualization). *Formula:* CEO Q&A uncertainty (%) = UnctWordsAnsCEO / WordsAnsCEO (DWZ 2021 Eqn 2). Uncertainty wordlist = LM 2011 Master Dictionary 'uncertainty' list (297 words, Aug 2014 version). CEO identified via Execucomp ceoann field. Used in body only as the dependent variable inside the CEO fixed-effect residualization that produces ClarityCEO and UncResCEO.	outputs/2_Textual_Analysis/2.2_Variables	Dzielínski et al. (2021) (Section 4.2, Eqn 2, p.13)
`UncPreCEO`	CEO-only presentation-segment uncertainty percentage (DWZ Eqn 1). *Formula:* CEO Presentation uncertainty (%) = UnctWordsPreCEO / WordsPreCEO. Uncertainty wordlist = LM 2011 (297 words).	outputs/2_Textual_Analysis/2.2_Variables	Dzielínski et al. (2021) (Section 4.2, Eqn 1, p.13)
`UncQue`	Analyst Q&A uncertainty percentage (DWZ Eqn 3). *Formula:* Analyst Q&A uncertainty (%) = UnctWordsQue / WordsQue. Uncertainty wordlist = LM 2011 (297 words). Used as ΔUncQue control in DWZ first-difference replication.	outputs/2_Textual_Analysis/2.2_Variables	Dzielínski et al. (2021) (Section 4.2, Eqn 3, p.13)

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Name	Description / Formula	Source	Reference
`UncResCEO`	CEO Q&A residual call-state uncertainty component (DWZ Eqn 4). *Formula:* $\widehat{\varepsilon}_{ic}$ = the residual from the panel regression UncAnsCEO$_{ic}$ = $\alpha + \sum_i \mathrm{FE}_i \cdot \mathbb{1}[\mathrm{CEO} = i] + \mathbf{X}_{ic}\boldsymbol{\gamma} + \varepsilon_{ic}$ that produces ClarityCEO. By OLS first-order conditions, the within-CEO mean of UncResCEO is exactly zero by construction. Captures call-level deviation in CEO Q&A uncertainty after stripping out the persistent CEO mean.	outputs/econometric/ceo_clarity_extended/	Dzieliński et al. (2021) (Section 4.4 Eqn 4, p.16)
`UncResCEO_QtrExp`	UncResCEO under within-tenure expanding window. *Formula:* As UncResCEO, but the residualization regression is re-estimated on an expanding within-CEO window. Strictly look-ahead-free.	outputs/econometric/ceo_clarity_expanding/	thesis QtrExp variant of Dzieliński et al. (2021) Eqn 4

A.3 Financial / Market Variables

Name	Description / Formula	Source	Reference
`BGTAvg_Amihud`	BGT 25-day symmetric average Amihud illiquidity (H7e). *Formula:* Mean daily Amihud illiquidity over [-25, +25] trading days around call (51-day symmetric window, day 0 INCLUDED). F1D extension to BGT 2018.	.../variables/bgt_long_window_amihud.py	bgt2018 (window); F1D extension (symmetric average shape)
`BGTAvg_Spread`	BGT 25-day symmetric average bid-ask spread (H14e). *Formula:* Mean daily relative bid-ask spread over [-25, +25] trading days around call (51-day symmetric window, day 0 INCLUDED). Spread = (ASK - BID) / ((ASK + BID) / 2). F1D extension to BGT 2018.	.../variables/bgt_long_window_spread.py	bgt2018 (window); F1D extension (symmetric average shape)
`BGTDelta_Amihud`	BGT 25-day post-pre Amihud illiquidity change (H7d). *Formula:* Mean Amihud over post window [+1, +25] minus mean over pre window [-25, -1] trading days (day 0 EXCLUDED). F1D extension to BGT 2018.	.../variables/bgt_long_window_amihud.py	bgt2018 (window); F1D extension (delta shape)
`BGTDelta_Spread`	BGT 25-day post-pre bid-ask spread change (H14d). *Formula:* Mean spread over [+1, +25] minus mean over [-25, -1] trading days (day 0 EXCLUDED). F1D extension to BGT 2018.	.../variables/bgt_long_window_spread.py	bgt2018 (window); F1D extension (delta shape)
`BGTLevel_Amihud`	Compute BGT (2018) long-window Amihud illiquidity around earnings calls. *Formula:* CRSP via get_raw_daily_data (see module: src/f1d/shared/variables/bgt_long_window_amihud.py)	CRSPviaget_raw_daily_data	Bushee, Gow & Taylor (2018, JAR) 56(1):85-121
`BGTLevel_Spread`	Compute BGT (2018) long-window closing bid-ask spread around earnings calls. *Formula:* CRSP via get_raw_daily_data (closing BID/ASK) (see module: src/f1d/shared/variables/bgt_long_window_spread.py)	CRSPviaget_raw_daily_data	Bushee, Gow & Taylor (2018, JAR) [window]; Lee (2016, TAR) [formula]
`Capex`	Build Capex = capxy_annual / atq_lag from raw Compustat quarterly data. *Formula:* Compustat/capxy_Q4/atq (see module: src/f1d/shared/variables/capex_intensity.py)	Compustat/capxy_Q4/atq	Bates, Kahle & Stulz (2009, JF)
`CashFlowAt`	Build CashFlow = oancfy / atq from raw Compustat quarterly data. *Formula:* Compustat/oancfy/atq (see module: src/f1d/shared/variables/cash_flow.py)	Compustat/oancfy/atq	Bates, Kahle & Stulz (2009, JF)

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Name	Description / Formula	Source	Reference
`CashRatio`	Build CashRatio = cheq / atq from raw Compustat quarterly data. *Formula:* Compustat/cheq/atq (see module: src/f1d/shared/variables/cash_holdings.py)	Compustat/cheq/atq	Bates et al. (2009) (Section II, p.1991; BKS use annual che/at — we use quarterly cheq/atq for panel alignment)
`ChangDebtChoice`	ChangDebtChoice (description not in module docstring) *Formula:* see module: src/f1d/shared/variables/_compustat_engine.py	.../variables/_compustat_engine.py	cdh2006
`ChangExternalFunding`	ChangExternalFunding (description not in module docstring) *Formula:* see module: src/f1d/shared/variables/_compustat_engine.py	.../variables/_compustat_engine.py	cdh2006
`DailyVola`	Build Volatility from raw CRSP daily stock files via CRSPEngine. *Formula:* CRSP/RET (see module: src/f1d/shared/variables/volatility.py)	CRSP/RET	dwz2021
`DebtToCapital`	Build DebtToCapital = total debt / (equity + total debt). *Formula:* Compustat/dlcq,dlttq,seqq (see module: src/f1d/shared/variables/debt_to_capital.py)	Compustat/dlcq,dlttq,seqq	rz1995
`DivDummy`	Build DivDummy = (dvy ¿ 0).astype(float) from raw Compustat quarterly data. *Formula:* Compustat/dvy¿0 (see module: src/f1d/shared/variables/dividend_payer.py)	Compustat/dvy¿0	Bates et al. (2009) (Section IV, p.2000; dividend payout dummy based on common dividends)
`DivPayerQ`	Build DivPayerQ = (dvpsxq ¿ 0).astype(float) from raw Compustat quarterly data. *Formula:* Compustat/dvpsxq¿0 (quarterly ex-date) (see module: src/f1d/shared/variables/dividend_payer_quarterly.py)	Compustat/dvpsxq¿0	Hoberg & Prabhala (2009, RFS) — Compustat item 26 (dvpsx) analog
`EarnVol`	Builder for Earnings Volatility (earnings_volatility) variable. *Formula:* Compustat (see module: src/f1d/shared/variables/earnings_volatility.py)	Compustat	Duong, Do, and Do (2025) (Appendix B, p.30; 3-year rolling std of qoq asset-scaled earnings)
`EquityDelayCon`	Hoberg-Maksimovic (2015) equity-market financing constraint index (H22). *Formula:* Hoberg-Maksimovic (2015) firm-year equity-market financing constraint measure. Higher values = more constrained firm. Panel column is lowercase 'equitydelaycon'; CamelCase name is the spec convention. See docs/provenance/h22.md for merge details.	inputs/Hoberg_Maksimovic/	hm2015
`FirmMat`	Builder for Firm Maturity (firm_maturity) variable. *Formula:* Compustat (see module: src/f1d/shared/variables/firm_maturity.py)	Compustat	Biddle, Hilary, and Verdi (2009) (Appendix A, p.130; years since first CRSP appearance); alt Leary and Roberts (2010) (App.˜A p.354)
`Leverage`	Build Leverage = (dlcq + dlttq) / atq (interest-bearing debt only). *Formula:* Compustat/dlcq,dlttq,atq (see module: src/f1d/shared/variables/book_lev.py)	Compustat/dlcq,dlttq,atq	bks2009, rz1995
`lnAssets`	Build Size = ln(atq) from raw Compustat quarterly data. *Formula:* Compustat/atq (see module: src/f1d/shared/variables/size.py)	Compustat/atq	dwz2021
`Loss`	Builder for Loss (H5 Control Variable). *Formula:* Compustat ibq ¡ 0 (see module: src/f1d/shared/variables/loss_dummy.py)	Compustatibq¡0	Biddle et al. (2009) (Appendix A, p.131; indicator for negative net income before extraordinary items)

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Name	Description / Formula	Source	Reference
PayoutRatio_q	Build PayoutRatio_q = (dvpspq × cshoq) / ibq from Compustat. *Formula:* Compustat/(dvpspq*cshoq)/ibq (quarterly) (see module: src/f1d/shared/variables/payout_ratio_quarterly.py)	Compustat/	dds2004
PRisk	Match quarterly PRisk onto each earnings call by (gvkey, calendar_quarter). *Formula:* Hassan et al. (2019) quarterly PRisk (see module: src/f1d/shared/variables/prisk_q.py)	Hassanetal.	hhlt2019
RDSales	Build RDSales = xrdq / atq from raw Compustat quarterly data. *Formula:* Compustat/xrdq,atq (see module: src/f1d/shared/variables/rd_intensity.py)	Compustat/xrdq,atq	bks2009, jjl2021
RepurchaseIntensity	Build RepurchaseIntensity = quarterly_prstkcy / lagged_atq from Compustat. *Formula:* Compustat/quarterly_prstkcy/lagged_atq (see module: src/f1d/shared/variables/repurchase_intensity.py)	Compustat/quarterly_prstkcy/lagged_atq	Thesis Advisor Suggestion
ROA	Build ROA = iby_annual / avg_assets from Compustat quarterly data. *Formula:* Compustat/iby,atq (see module: src/f1d/shared/variables/roa.py)	Compustat/iby,atq	Wang (2020), dwz2021
SalesGrowth	Build SalesGrowth from raw Compustat quarterly data (annual Q4 panel). *Formula:* Compustat/saley_Q4_YoY (see module: src/f1d/shared/variables/sales_growth.py)	Compustat/saley_Q4_YoY	Biddle et al. (2009) (Section 3.2, p.117; pct change in sales)
sCFO	Build sCFO = rolling 5-year std (min 3 yrs) of (oancfy/atq_{t-1}) per gvkey. *Formula:* Compustat/oancfy/atq_{t-1} rolling std (see module: src/f1d/shared/variables/ocf_volatility.py)	Compustat/oancfy/atq_{t-1}rollingstd	Biddle et al. (2009) (Appendix A, p.130; 5-year rolling std of CFO/avg assets)
StockPrice	Compute stock price at earnings call date (VECTORIZED). *Formula:* CRSP via get_raw_daily_data (see module: src/f1d/shared/variables/stock_price.py)	CRSPviaget_raw_daily_data	Amiram, Owens, and Rozenbaum (2016) (Appendix A, p.137; daily CRSP stock price; justification cites Stoll 1978)
SurpDec	Signed-quintile earnings surprise rank (-5 to +5) within calendar quarter. *Formula:* Signed quintile rank of percentage earnings surprise per DWZ 2021 Appendix Table A.1: raw surprise = (actual - consensus forecast earnings) / share price 5 trading days before announcement, x 100. Firms grouped into 5 bins of positive surprise (+5 largest through +1 smallest positive), 0 for zero surprises, and 5 bins of negative surprise (-1 smallest negative through -5 largest negative). DWZ footnote 32 notes this decile convention follows Hirshleifer, Lim & Teoh (2009) and DellaVigna & Pollet (2009).	.../variables/earnings_surprise.py	Dzieliński et al. (2021) (Appendix Table A.1, p.55; convention cites DellaVigna-Pollet 2009 + Hirshleifer-Lim-Teoh 2009)
TobinsQ	Build TobinsQ = (AT + cshoq*prccq - CEQ) / AT from raw Compustat quarterly data. *Formula:* Compustat/(atq+cshoq*prccq-ceqq)/atq (see module: src/f1d/shared/variables/tobins_q.py)	Compustat/	Bates, Kahle & Stulz (2009, JF)
Turnover	Compute share turnover at earnings call date (VECTORIZED). *Formula:* CRSP via get_raw_daily_data (see module: src/f1d/shared/variables/turnover.py)	CRSPviaget_raw_daily_data	Amihud (2002) (Section 2.1, p.33; alt rolling-window def in Chang, Dasgupta, and Hilary (2006) Appendix B p.3045)

A.4 Econometric / Indicator Variables

Name	Description / Formula	Source	Reference
`BelowIG`	Below-IG credit rating subsample indicator (H1.2). *Formula:* Binary = 1 if firm's S&P long-term issuer credit rating is below investment grade (BB+ and below) at call date; 0 otherwise. Rated firms only (Unrated firms get separate indicator).	runtime	thesis (H1.2 rating subsample)
`HighCFvol`	High cash-flow-volatility moderator (H1.3). *Formula:* Binary = 1 if firm-quarter Han-Qiu (2007) cash-flow-volatility (16-quarter rolling std of OCF / |mean| of OCF) is above the within-industry-year median; 0 otherwise.	runtime	Han and Qiu (2007); thesis (H1.3 split convention)
`HighTSIMM`	High product-market-similarity indicator (H1.1). *Formula:* Binary = 1 if firm-year TotalSimilarity (HP 2016 product market similarity) is above the within-year median; 0 otherwise.	runtime	hp2016 (TSIMM construction); thesis (H1.1 split convention)
`Lagged_DV`	Generic lagged-DV control (1Q lag of suite-specific DV). *Formula:* 1-quarter lag of the dependent variable, constructed at runtime per spec (DV varies per suite). Always included as base control to absorb persistence.	runtime	thesis convention (per feedback_lagged_dv.md)
`Unrated`	No-rating subsample indicator (H1.2). *Formula:* Binary = 1 if firm has no S&P rating at call date; 0 otherwise.	runtime	thesis (H1.2 rating subsample)
`z_log_TotalSimilarity`	z-scored log of HP 2016 TotalSimilarity (continuous moderator, H1.1). *Formula:* z-score of log(1 + TotalSimilarity) computed within year. TotalSimilarity = HP 2016 firm-year product market similarity score (sum of pairwise cosine similarities $\times$ 100).	runtime	hp2016

A.5 Runtime Derivatives (Lead/Lag/Centered/Interaction)

Name	Description / Formula	Source	Reference
`AbsSurpDec`	Absolute earnings surprise quintile magnitude. *Formula:* abs(SurpDec) — unsigned magnitude of signed-quintile earnings surprise rank, values 0-5.	runtime	thesis (magnitude control independent of sign)
`BGTAvg_Amihud_lead1`	1Q lead of BGTAvg_Amihud. *Formula:* 1-quarter lead of BGTAvg_Amihud	runtime	bgt2018 (window); F1D extension (symmetric average shape)
`BGTAvg_Spread_lead1`	1Q lead of BGTAvg_Spread. *Formula:* 1-quarter lead of BGTAvg_Spread	runtime	bgt2018 (window); F1D extension (symmetric average shape)
`BGTDelta_Amihud_lead1`	1Q lead of BGTDelta_Amihud. *Formula:* 1-quarter lead of BGTDelta_Amihud	runtime	bgt2018 (window); F1D extension (delta shape)
`BGTDelta_Spread_lead1`	1Q lead of BGTDelta_Spread. *Formula:* 1-quarter lead of BGTDelta_Spread	runtime	bgt2018 (window); F1D extension (delta shape)
`BGTLevel_Amihud_lead1`	1Q lead of BGTLevel_Amihud. *Formula:* 1-quarter lead of BGTLevel_Amihud	runtime	bgt2018

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Name	Description / Formula	Source	Reference
BGTLevel_Spread_lead1	1Q lead of BGTLevel_Spread. *Formula:* 1-quarter lead of BGTLevel_Spread	runtime	bgt2018, lee2016
Capex_lead	1Q lead of Capex. *Formula:* 1-quarter lead of Capex	runtime	Bates, Kahle & Stulz (2009, JF)
Capex_lead2	2Q lead of Capex. *Formula:* 2-quarter lead of Capex	runtime	Bates, Kahle & Stulz (2009, JF)
Capex_lead3	3Q lead of Capex. *Formula:* 3-quarter lead of Capex	runtime	Bates, Kahle & Stulz (2009, JF)
Capex_lead4	4Q lead of Capex. *Formula:* 4-quarter lead of Capex	runtime	Bates, Kahle & Stulz (2009, JF)
CashRatio_lead	1Q lead of CashRatio. *Formula:* 1-quarter lead of CashRatio	runtime	see base var
ChangDebtChoice_lead	1Q lead of ChangDebtChoice. *Formula:* 1-quarter lead of ChangDebtChoice	runtime	cdh2006
ChangExternalFunding_lead	1Q lead of ChangExternalFunding. *Formula:* 1-quarter lead of ChangExternalFunding	runtime	cdh2006
DebtToCapital_lead	1Q lead of DebtToCapital. *Formula:* 1-quarter lead of DebtToCapital	runtime	rz1995
DeltaILLIQ_lead1	1Q lead of DeltaILLIQ. *Formula:* 1-quarter lead of DeltaILLIQ	runtime	amihud02
DISP_lag	1Q lag of DISP. *Formula:* 1-quarter lag of DISP	runtime	Wang (2020)
DISP_lead	1Q lead of DISP. *Formula:* 1-quarter lead of DISP	runtime	Wang (2020)
DivPayerQ_lead1	1Q lead of DivPayerQ. *Formula:* 1-quarter lead of DivPayerQ	runtime	Hoberg & Prabhala (2009, RFS) – primary; Chetty & Saez (2005, QJE) – secondary (firm-quarter frequency precedent)
DSPREAD_lead1	1Q lead of DSPREAD. *Formula:* 1-quarter lead of DSPREAD	runtime	lee2016
EquityDelayCon_lead	1Q lead of EquityDelayCon. *Formula:* 1-quarter lead of EquityDelayCon	runtime	hm2015
Leverage_lead	1Q lead of Leverage. *Formula:* 1-quarter lead of Leverage	runtime	bks2009, rz1995

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Name	Description / Formula	Source	Reference
PayoutRatio_q_lead_qtr	1Q lead of PayoutRatio_q (qtr-aligned). *Formula:* 1-quarter lead of PayoutRatio_q (calendar-quarter aligned)	runtime	dds2004
PostCallAmihud	Panel-time post-call Amihud illiquidity level (H7b). *Formula:* Panel-time mean daily Amihud illiquidity over the post-call period (call quarter + N following quarters per H7b runner spec). Computed on panel structure with two-way clustering.	runtime	thesis (H7b panel-time post-call construction)
PostCallAmihud_lead1	1Q lead of PostCallAmihud. *Formula:* 1-quarter lead of PostCallAmihud	runtime	thesis (H7b panel-time post-call construction)
PostCallSpread	Panel-time post-call bid-ask spread level (H14b). *Formula:* Panel-time mean daily relative bid-ask spread over the post-call period (call quarter + N following quarters per H14b runner spec). Computed on panel structure with two-way clustering.	runtime	thesis (H14b panel-time post-call construction)
PostCallSpread_lead1	1Q lead of PostCallSpread. *Formula:* 1-quarter lead of PostCallSpread	runtime	thesis (H14b panel-time post-call construction)
PRisk_lag	1Q lag of PRisk. *Formula:* 1-quarter lag of PRisk	runtime	hhlt2019
PRisk_lag2	2Q lag of PRisk. *Formula:* 2-quarter lag of PRisk	runtime	hhlt2019
RDSales_lead	1Q lead of RDSales. *Formula:* 1-quarter lead of RDSales	runtime	bks2009, jjl2021
RepurchaseIntensity_lead_qtr	1Q lead of RepurchaseIntensity (qtr-aligned). *Formula:* 1-quarter lead of RepurchaseIntensity (calendar-quarter aligned)	runtime	Thesis Advisor Suggestion
ClarityCEO_c	Mean-centered ClarityCEO. *Formula:* ClarityCEO minus its sample mean (mean-centered for interaction interpretability)	runtime	thesis (interaction interpretability)
ClarityCEO_QtrExp_c	Mean-centered ClarityCEO_QtrExp. *Formula:* ClarityCEO_QtrExp minus its sample mean	runtime	thesis (interaction interpretability)
Post	DiD post-event-period indicator (Phase E sudden-death). *Formula:* Binary = 1 if call-quarter is on or after the death-event quarter for the treated firm (and on or after the matched event-quarter for the matched-control firm); 0 otherwise.	runtime	Ghafoor, Yousaf, and Li (2023)
Treated	DiD treated-firm indicator (Phase E sudden-death). *Formula:* Binary = 1 if firm experienced a CEO sudden-death event in 2002–2018; 0 if PSM-matched control firm.	runtime	Bennedsen, Pérez-González, and Wolfenzon (2020); Ghafoor et al. (2023)
Treated_x_Post	DiD ATT term (Phase E sudden-death). *Formula:* Treated $\times$ Post. The coefficient on this product is the average treatment effect on the treated.	runtime	Bertrand, Duflo, and Mullainathan (2004); Ghafoor et al. (2023)

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Name	Description / Formula	Source	Reference
UncPreCEO_c	Mean-centered UncPreCEO. *Formula:* UncPreCEO minus its sample mean (mean-centered for interaction interpretability)	runtime	thesis (interaction interpretability)
UncPreCEO_c_x_HighCFvol	Interaction: UncPreCEO_c × HighCFvol (H1.3). *Formula:* Product: UncPreCEO_c * HighCFvol	runtime	thesis (CFvol moderator interaction)
UncPreCEO_c_x_Unrated	Interaction: UncPreCEO_c × Unrated (HFC). *Formula:* Product: UncPreCEO_c * Unrated	runtime	thesis (HFC interaction)
UncResCEO_c	Mean-centered UncResCEO. *Formula:* UncResCEO minus its sample mean (mean-centered for interaction interpretability)	runtime	thesis (interaction interpretability)
UncResCEO_c_x_HighCFvol	Interaction: UncResCEO_c × HighCFvol (H1.3). *Formula:* Product: UncResCEO_c * HighCFvol	runtime	thesis (CFvol moderator interaction)
UncResCEO_c_x_Unrated	Interaction: UncResCEO_c × Unrated (HFC). *Formula:* Product: UncResCEO_c * Unrated	runtime	thesis (HFC interaction)
UncResCEO_QtrExp_c	Mean-centered UncResCEO_QtrExp. *Formula:* UncResCEO_QtrExp minus its sample mean	runtime	thesis (interaction interpretability)
UncResCEO_QtrExp_c_x_Unrated	Interaction: UncResCEO_QtrExp_c × Unrated (HFC QtrExp). *Formula:* Product: UncResCEO_QtrExp_c * Unrated	runtime	thesis (HFC QtrExp interaction)

A.6 Stage 9

Name	Description / Formula	Source	Reference
CCCL	SEC comment letter received (binary, forward-looking) *Formula:* 1 if SEC comment letter received in next quarter, 0 otherwise	inputs/EDGAR_CommentLetters	cdm2013
DeltaILLIQ	Change in Amihud illiquidity around earnings call *Formula:* PostCallILLIQ - PreCallILLIQ over [+1,+3] vs [-3,-1] days	inputs/CRSP_DSF	amihud02
DISP	Analyst forecast dispersion (Wang 2020) *Formula:* SD(latest analyst EPS forecasts in T-31..T-1 window) / prccq_prior	inputs/tr_ibes	Wang (2020)
DSPREAD	Change in closing bid-ask spread around earnings call *Formula:* mean(RelSpread[+1,+3]) - mean(RelSpread[-3,-1]) where RelSpread = 2*(ASK-BID)/(ASK+BID)	inputs/CRSP_DSF	lee2016
GEPU_log	Davis (2016, NBER WP 22740) Global Economic Policy Uncertainty (GEPU) index (log-transformed, current-$ GDP weights) *Formula:* log(GEPU_current) — GDP-weighted average of 16 country EPU indices, matched by calendar month of call start_date	inputs/EconomicPolicyUncertaintyIndex/Global	Davis (2016)

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Name	Description / Formula	Source	Reference
`GPR_log`	Caldara & Iacoviello (2022, AER) headline Geopolitical Risk index (log-transformed, 10-newspaper recent version) *Formula:* log(GPR) — recent headline Geopolitical Risk index, matched by calendar month of call start_date	inputs/matteoiacoviello/GPR_Global	ci2022
`SEC_Letters_fwd`	SEC comment letter count (forward-looking) *Formula:* Count of SEC-originated letters (UPLOAD type) in next calendar quarter	inputs/EDGAR_CommentLetters	cdm2013 (thesis extension to count)
`TotalSimilarity`	Hoberg-Phillips total product similarity *Formula:* Sum of pairwise cosine similarities from 10-K product descriptions	inputs/TNIC	hp2016
`US_EPU_log`	Baker, Bloom & Davis (2016, QJE) US News-Based Economic Policy Uncertainty index (log-transformed) *Formula:* log(News_Based_Policy_Uncert_Index) matched by calendar month of call start_date	inputs/EconomicPolicyUncertaintyIndex/US	bbd2016